



StataNow 19.5  
BE-Basic Edition

# Statistics and Data Science

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NBU

## Notes:

1. Unicode is supported; see [help unicode advice](#).
2. New update available; type `-update all-`

```
1 . do "/var/folders/vn/xyfk53kn6yzb3ch5v45030kc0000gn/T//SD37829.000000"

2 . *****
3 . * Brexit paper
4 . * Ground-zero Stata replication from panel_wb.xlsx
5 . * Current starting file has 468 observations
6 . * 52 countries x 9 years, 2016–2024
7 . *****
8 .
9 . clear all

10 . set more off

11 .
12 . *****
13 . * 1. Import starting file
14 . *****
15 .
16 . import excel "panel_wb.xlsx", sheet("Sheet 1") firstrow clear
    (20 vars, 468 obs)

17 .
18 . describe
```

Contains data  
Observations: 468  
Variables: 20

| Variable name   | Storage type | Display format | Value label | Variable label                  |
|-----------------|--------------|----------------|-------------|---------------------------------|
| Year            | int          | %10.0g         |             | Year                            |
| Domicile_name~y | str28        | %28s           |             | Domicile_named_country          |
| Applicants      | long         | %10.0g         |             | Applicants                      |
| Accepted_appl~s | int          | %10.0g         |             | Accepted_applicants             |
| Acceptance_rate | str19        | %19s           |             | Acceptance_rate                 |
| Applicants_wi~s | int          | %10.0g         |             | Applicants_with_no_applications |
| log_app         | double       | %10.0g         |             | log_app                         |
| EU              | byte         | %10.0g         |             | EU                              |
| Treated         | byte         | %10.0g         |             | Treated                         |
| Post            | byte         | %10.0g         |             | Post                            |
| event_time      | byte         | %10.0g         |             | event_time                      |
| EU_group        | str12        | %12s           |             | EU_group                        |
| treated_factor  | str23        | %23s           |             | treated_factor                  |
| regression_we~t | double       | %10.0g         |             | regression_weight               |
| did             | byte         | %10.0g         |             | did                             |
| iso3c           | str3         | %9s            |             | iso3c                           |
| GDPpc           | double       | %10.0g         |             | GDPpc                           |
| Pop15_64        | long         | %10.0g         |             | Pop15_64                        |
| YouthUnempl     | double       | %10.0g         |             | YouthUnempl                     |
| EduSpendGDP     | double       | %10.0g         |             | EduSpendGDP                     |

Sorted by:

Note: Dataset has changed since last saved.

```
19 . count
    468

20 . tab Year
```

| Year  | Freq. | Percent | Cum.   |
|-------|-------|---------|--------|
| 2016  | 52    | 11.11   | 11.11  |
| 2017  | 52    | 11.11   | 22.22  |
| 2018  | 52    | 11.11   | 33.33  |
| 2019  | 52    | 11.11   | 44.44  |
| 2020  | 52    | 11.11   | 55.56  |
| 2021  | 52    | 11.11   | 66.67  |
| 2022  | 52    | 11.11   | 77.78  |
| 2023  | 52    | 11.11   | 88.89  |
| 2024  | 52    | 11.11   | 100.00 |
| Total | 468   | 100.00  |        |

```

21 .
22 . *****
23 . * 2. Panel identifier and basic checks
24 . *****
25 .
26 . capture drop DMC

27 . encode Domicile_named_country, gen(DMC)

28 .
29 . xtset DMC Year

    Panel variable: DMC (strongly balanced)
    Time variable: Year, 2016 to 2024
    Delta: 1 unit

30 . isid DMC Year

31 .
32 . assert Post == (Year >= 2021)

33 . assert did == Treated*Post

34 .
35 . count if missing(log_app, regression_weight, did, GDPpc, Pop15_64, YouthUnempl)
    0

36 . summ regression_weight

    Variable |      Obs      Mean   Std. dev.   Min      Max
    -----+-----
 regression~t |      468         1   .7706347   6.65e-11   4.177701

37 .
38 . *****
39 . * 3. Recreate the R transformations exactly
40 . *   Unweighted centering over the 468-row estimation sample
41 . *****
42 .
43 . capture drop log_GDPpc c_log_GDPpc

44 . gen double log_GDPpc = ln(GDPpc)

45 . quietly summarize log_GDPpc, meanonly

46 . gen double c_log_GDPpc = log_GDPpc - r(mean)

47 .
48 . capture drop log_Pop15_64 c_log_Pop15_64

49 . gen double log_Pop15_64 = ln(Pop15_64)

50 . quietly summarize log_Pop15_64, meanonly

51 . gen double c_log_Pop15_64 = log_Pop15_64 - r(mean)

52 .
53 . capture drop c_YouthUnempl

54 . quietly summarize YouthUnempl, meanonly

55 . gen double c_YouthUnempl = YouthUnempl - r(mean)

56 .
57 . summ c_log_GDPpc c_log_Pop15_64 c_YouthUnempl

```

| Variable      | Obs | Mean      | Std. dev. | Min       | Max      |
|---------------|-----|-----------|-----------|-----------|----------|
| c_log_GDPpc   | 468 | -1.29e-16 | .7805845  | -2.682964 | 1.301677 |
| c_log_Pop~64  | 468 | 5.24e-16  | 1.731308  | -3.282579 | 4.799478 |
| c_YouthUnempl | 468 | 3.74e-16  | 8.43441   | -15.02041 | 31.90759 |

```

58 .
59 . *****
60 . * 4. Model V interaction terms
61 . *****
62 .
63 . capture drop did_GDP did_Pop did_YU

64 . gen double did_GDP = did*c_log_GDPpc

65 . gen double did_Pop = did*c_log_Pop15_64

66 . gen double did_YU = did*c_YouthUnempl

67 .
68 . *****
69 . * 5. Explicit fixed-effect dummies
70 . *   We use pooled xtscd with explicit country and year FE
71 . *   so the coefficient objective is weighted LSDV
72 . *****
73 .
74 . capture drop FE_C*

75 . capture drop FE_Y*

76 .
77 . tab DMC, gen(FE_C)

```

| Domicile_named_country    | Freq. | Percent | Cum.  |
|---------------------------|-------|---------|-------|
| Australia                 | 9     | 1.92    | 1.92  |
| Austria                   | 9     | 1.92    | 3.85  |
| Belgium                   | 9     | 1.92    | 5.77  |
| Botswana                  | 9     | 1.92    | 7.69  |
| Brunei Darussalam         | 9     | 1.92    | 9.62  |
| Bulgaria                  | 9     | 1.92    | 11.54 |
| China                     | 9     | 1.92    | 13.46 |
| Croatia                   | 9     | 1.92    | 15.38 |
| Cyprus (European Union)   | 9     | 1.92    | 17.31 |
| Czech Republic            | 9     | 1.92    | 19.23 |
| Denmark                   | 9     | 1.92    | 21.15 |
| Egypt                     | 9     | 1.92    | 23.08 |
| Estonia                   | 9     | 1.92    | 25.00 |
| Finland                   | 9     | 1.92    | 26.92 |
| France                    | 9     | 1.92    | 28.85 |
| Germany                   | 9     | 1.92    | 30.77 |
| Greece                    | 9     | 1.92    | 32.69 |
| Hungary                   | 9     | 1.92    | 34.62 |
| Indonesia                 | 9     | 1.92    | 36.54 |
| Iran, Islamic Republic of | 9     | 1.92    | 38.46 |
| Ireland                   | 9     | 1.92    | 40.38 |
| Israel                    | 9     | 1.92    | 42.31 |
| Italy                     | 9     | 1.92    | 44.23 |
| Korea, Republic of        | 9     | 1.92    | 46.15 |
| Latvia                    | 9     | 1.92    | 48.08 |
| Lebanon                   | 9     | 1.92    | 50.00 |
| Lithuania                 | 9     | 1.92    | 51.92 |
| Luxembourg                | 9     | 1.92    | 53.85 |
| Malaysia                  | 9     | 1.92    | 55.77 |
| Malta                     | 9     | 1.92    | 57.69 |
| Mauritius                 | 9     | 1.92    | 59.62 |
| Mexico                    | 9     | 1.92    | 61.54 |
| Netherlands               | 9     | 1.92    | 63.46 |
| New Zealand               | 9     | 1.92    | 65.38 |
| Norway                    | 9     | 1.92    | 67.31 |
| Poland                    | 9     | 1.92    | 69.23 |
| Portugal                  | 9     | 1.92    | 71.15 |
| Qatar                     | 9     | 1.92    | 73.08 |
| Romania                   | 9     | 1.92    | 75.00 |
| Russian Federation        | 9     | 1.92    | 76.92 |
| Slovakia                  | 9     | 1.92    | 78.85 |
| Slovenia                  | 9     | 1.92    | 80.77 |
| Spain                     | 9     | 1.92    | 82.69 |
| Sweden                    | 9     | 1.92    | 84.62 |

|                              |     |        |        |
|------------------------------|-----|--------|--------|
| Switzerland                  | 9   | 1.92   | 86.54  |
| Tanzania, United Republic of | 9   | 1.92   | 88.46  |
| Trinidad and Tobago          | 9   | 1.92   | 90.38  |
| Turkey                       | 9   | 1.92   | 92.31  |
| Uganda                       | 9   | 1.92   | 94.23  |
| United Arab Emirates         | 9   | 1.92   | 96.15  |
| United States of America     | 9   | 1.92   | 98.08  |
| Zimbabwe                     | 9   | 1.92   | 100.00 |
| Total                        | 468 | 100.00 |        |

```
78 . tab Year, gen(FE_Y)
```

| Year  | Freq. | Percent | Cum.   |
|-------|-------|---------|--------|
| 2016  | 52    | 11.11   | 11.11  |
| 2017  | 52    | 11.11   | 22.22  |
| 2018  | 52    | 11.11   | 33.33  |
| 2019  | 52    | 11.11   | 44.44  |
| 2020  | 52    | 11.11   | 55.56  |
| 2021  | 52    | 11.11   | 66.67  |
| 2022  | 52    | 11.11   | 77.78  |
| 2023  | 52    | 11.11   | 88.89  |
| 2024  | 52    | 11.11   | 100.00 |
| Total | 468   | 100.00  |        |

```
79 .
80 . * FE_C1 and FE_Y1 are reference categories
81 . * The current file has exactly 52 countries and 9 years
82 .
83 . *****
84 . * 6. Install commands if needed
85 . *****
86 .
87 . capture which xtsc
88 . if _rc != 0 {
89 .     ssc install xtsc
90 . }
91 .
92 . capture which esttab
93 . if _rc != 0 {
94 .     ssc install estout
95 . }
96 .
97 . *****
98 . * 7. Table 3 models II-V
99 . *
100 . * Model I in the manuscript uses 520 observations,
101 . * 2016-2025. It cannot be reproduced from this 468-row
102 . * 2016-2024 starting file.
103 . *****
104 .
105 . estimates clear
106 .
107 . * Model II
108 . xtsc log_app did c_log_GDPpc FE_C2-FE_C52 FE_Y2-FE_Y9 [aw=regression_weight], pooled lag(1)
```

```
Regression with Driscoll-Kraay standard errors   Number of obs   =       468
Method: Pooled OLS                             Number of groups =        52
Group variable (i): DMC                        F( 61,      8)   =    1621.60
maximum lag: 1                                Prob > F         =     0.0000
                                             R-squared        =     0.9634
                                             Root MSE        =     0.2426
```

| log_app     | Drisc/Kraay |           |       |       |           | [95% conf. interval] |
|-------------|-------------|-----------|-------|-------|-----------|----------------------|
|             | Coefficient | std. err. | t     | P> t  |           |                      |
| did         | -1.048992   | .1105316  | -9.49 | 0.000 | -1.303878 | -.7941058            |
| c_log_GDPpc | -.7269999   | .2561376  | -2.84 | 0.022 | -1.317654 | -.1363455            |
| FE_C2       | .1919747    | .0837136  | 2.29  | 0.051 | -.0010693 | .3850187             |

|        |           |          |        |       |           |           |
|--------|-----------|----------|--------|-------|-----------|-----------|
| FE_C3  | .7152041  | .0693148 | 10.32  | 0.000 | .5553639  | .8750443  |
| FE_C4  | -2.236313 | .3069413 | -7.29  | 0.000 | -2.944121 | -1.528505 |
| FE_C5  | -.3800999 | .1344119 | -2.83  | 0.022 | -.6900542 | -.0701455 |
| FE_C6  | .34543    | .2880433 | 1.20   | 0.265 | -.318799  | 1.009659  |
| FE_C7  | 2.758882  | .3797063 | 7.27   | 0.000 | 1.883278  | 3.634486  |
| FE_C8  | -1.207505 | .2127324 | -5.68  | 0.000 | -1.698067 | -.7169429 |
| FE_C9  | 1.371042  | .0310475 | 44.16  | 0.000 | 1.299447  | 1.442638  |
| FE_C10 | .3538777  | .1019315 | 3.47   | 0.008 | .1188231  | .5889323  |
| FE_C11 | -.1397827 | .070117  | -1.99  | 0.081 | -.3014729 | .0219074  |
| FE_C12 | -.7404276 | .3691404 | -2.01  | 0.080 | -1.591667 | .1108118  |
| FE_C13 | -.8905244 | .1759892 | -5.06  | 0.001 | -1.296356 | -.4846925 |
| FE_C14 | -.0570813 | .2266152 | -0.25  | 0.807 | -.5796569 | .4654942  |
| FE_C15 | 2.225306  | .0644206 | 34.54  | 0.000 | 2.076751  | 2.37386   |
| FE_C16 | 1.868326  | .0541422 | 34.51  | 0.000 | 1.743474  | 1.993178  |
| FE_C17 | .9877937  | .1250171 | 7.90   | 0.000 | .6995039  | 1.276084  |
| FE_C18 | .0589708  | .1499533 | 0.39   | 0.704 | -.2868221 | .4047637  |
| FE_C19 | -.9135675 | .4222009 | -2.16  | 0.062 | -1.887165 | .0600296  |
| FE_C20 | -1.585682 | .3016639 | -5.26  | 0.001 | -2.28132  | -.8900439 |
| FE_C21 | 2.391435  | .143174  | 16.70  | 0.000 | 2.061276  | 2.721595  |
| FE_C22 | -1.636242 | .0677132 | -24.16 | 0.000 | -1.792389 | -1.480095 |
| FE_C23 | 1.874644  | .0356069 | 52.65  | 0.000 | 1.792535  | 1.956754  |
| FE_C24 | .7522959  | .0939975 | 8.00   | 0.000 | .5355374  | .9690545  |
| FE_C25 | -.6565857 | .226636  | -2.90  | 0.020 | -1.179209 | -.1339621 |
| FE_C26 | -1.599243 | .4611711 | -3.47  | 0.008 | -2.662705 | -.5357803 |
| FE_C27 | .3599609  | .365952  | 0.98   | 0.354 | -.4839261 | 1.203848  |
| FE_C28 | .4054509  | .2363543 | 1.72   | 0.125 | -.1395831 | .9504848  |
| FE_C29 | 1.44309   | .151385  | 9.53   | 0.000 | 1.093995  | 1.792184  |
| FE_C30 | -1.398526 | .1470324 | -9.51  | 0.000 | -1.737583 | -1.059468 |
| FE_C31 | -1.284487 | .2002103 | -6.42  | 0.000 | -1.746173 | -.822801  |
| FE_C32 | -1.66096  | .2745658 | -6.05  | 0.000 | -2.29411  | -1.027811 |
| FE_C33 | .7011066  | .1026611 | 6.83   | 0.000 | .4643696  | .9378436  |
| FE_C34 | -1.290457 | .0393634 | -32.78 | 0.000 | -1.38123  | -1.199685 |
| FE_C35 | .9696855  | .2020861 | 4.80   | 0.001 | .5036741  | 1.435697  |
| FE_C36 | 1.303793  | .2299189 | 5.67   | 0.000 | .7735988  | 1.833987  |
| FE_C37 | .9625008  | .2606079 | 3.69   | 0.006 | .361538   | 1.563464  |
| FE_C38 | .5474742  | .1757527 | 3.12   | 0.014 | .1421877  | .9527606  |
| FE_C39 | 1.046688  | .2461084 | 4.25   | 0.003 | .4791609  | 1.614215  |
| FE_C40 | -.0918007 | .1549631 | -0.59  | 0.570 | -.4491462 | .2655447  |
| FE_C41 | -.2537223 | .217297  | -1.17  | 0.277 | -.7548101 | .2473654  |
| FE_C42 | -1.778376 | .0994939 | -17.87 | 0.000 | -2.007809 | -1.548942 |
| FE_C43 | 1.85559   | .1191851 | 15.57  | 0.000 | 1.580749  | 2.130431  |
| FE_C44 | .7457307  | .041493  | 17.97  | 0.000 | .6500475  | .8414138  |
| FE_C45 | .9438002  | .1022007 | 9.23   | 0.000 | .7081249  | 1.179475  |
| FE_C46 | -3.483651 | .7097344 | -4.91  | 0.001 | -5.120301 | -1.847    |
| FE_C47 | -1.79919  | .1965816 | -9.15  | 0.000 | -2.252508 | -1.345872 |
| FE_C48 | .3417135  | .2182211 | 1.57   | 0.156 | -.1615053 | .8449322  |
| FE_C49 | -3.267392 | .7995233 | -4.09  | 0.004 | -5.111096 | -1.423688 |
| FE_C50 | 1.554689  | .0761213 | 20.42  | 0.000 | 1.379153  | 1.730226  |
| FE_C51 | 2.279498  | .0491642 | 46.36  | 0.000 | 2.166125  | 2.392871  |
| FE_C52 | -1.887051 | .6635442 | -2.84  | 0.022 | -3.417187 | -.3569153 |
| FE_Y2  | -.02937   | .0061967 | -4.74  | 0.001 | -.0436597 | -.0150803 |
| FE_Y3  | .035244   | .012685  | 2.78   | 0.024 | .0059923  | .0644958  |
| FE_Y4  | .038151   | .017559  | 2.17   | 0.062 | -.0023402 | .0786422  |
| FE_Y5  | -.0146375 | .0047242 | -3.10  | 0.015 | -.0255316 | -.0037434 |
| FE_Y6  | .1856785  | .067382  | 2.76   | 0.025 | .0302953  | .3410617  |
| FE_Y7  | .0478769  | .073229  | 0.65   | 0.532 | -.1209896 | .2167434  |
| FE_Y8  | .0385807  | .0758036 | 0.51   | 0.625 | -.1362227 | .2133841  |
| FE_Y9  | .0260922  | .0790422 | 0.33   | 0.750 | -.1561796 | .2083639  |
| _cons  | 6.810033  | .1167786 | 58.32  | 0.000 | 6.540741  | 7.079325  |

109 . estimates store M2

110 .

111 . \* Model III

112 . xtsccl log\_app did c\_log\_GDPpc c\_log\_Pop15\_64 FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight], pooled lag(1)

|                                                |                  |   |        |
|------------------------------------------------|------------------|---|--------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468    |
| Method: Pooled OLS                             | Number of groups | = | 52     |
| Group variable (i): DMC                        | F( 62, 8)        | = | 751.04 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000 |
|                                                | R-squared        | = | 0.9685 |
|                                                | Root MSE         | = | 0.2253 |

| log_app | Coefficient | Drisc/Kraay<br>std. err. | t     | P> t  | [95% conf. interval] |
|---------|-------------|--------------------------|-------|-------|----------------------|
| did     | -.9800599   | .1007547                 | -9.73 | 0.000 | -1.212401 - .7477191 |

|                |           |          |       |       |           |           |
|----------------|-----------|----------|-------|-------|-----------|-----------|
| c_log_GDPpc    | -.4591213 | .2593755 | -1.77 | 0.115 | -1.057242 | .1389997  |
| c_log_Pop15_64 | 3.07453   | .4600657 | 6.68  | 0.000 | 2.013616  | 4.135443  |
| FE_C2          | 3.31677   | .5320777 | 6.23  | 0.000 | 2.089797  | 4.543743  |
| FE_C3          | 3.171271  | .4213934 | 7.53  | 0.000 | 2.199536  | 4.143006  |
| FE_C4          | 5.493615  | 1.092866 | 5.03  | 0.001 | 2.973461  | 8.013769  |
| FE_C5          | 11.65472  | 1.778603 | 6.55  | 0.000 | 7.553253  | 15.75619  |
| FE_C6          | 4.705498  | .5433775 | 8.66  | 0.000 | 3.452467  | 5.958529  |
| FE_C7          | -9.476662 | 1.828878 | -5.18 | 0.001 | -13.69406 | -5.259261 |
| FE_C8          | 4.694359  | .8120309 | 5.78  | 0.000 | 2.821813  | 6.566906  |
| FE_C9          | 10.32658  | 1.322809 | 7.81  | 0.000 | 7.276177  | 13.37698  |
| FE_C10         | 3.111961  | .3850113 | 8.08  | 0.000 | 2.224123  | 3.999799  |
| FE_C11         | 4.394292  | .6721917 | 6.54  | 0.000 | 2.844215  | 5.944369  |
| FE_C12         | -4.68901  | .8013285 | -5.85 | 0.000 | -6.536876 | -2.841143 |
| FE_C13         | 8.319771  | 1.277287 | 6.51  | 0.000 | 5.374341  | 11.2652   |
| FE_C14         | 4.777324  | .6180384 | 7.73  | 0.000 | 3.352125  | 6.202523  |
| FE_C15         | -.6103679 | .3825429 | -1.60 | 0.149 | -1.492513 | .2717775  |
| FE_C16         | -1.76008  | .4895811 | -3.60 | 0.007 | -2.889056 | -.6311042 |
| FE_C17         | 3.892519  | .4837832 | 8.05  | 0.000 | 2.776913  | 5.008125  |
| FE_C18         | 3.120359  | .4627855 | 6.74  | 0.000 | 2.053173  | 4.187544  |
| FE_C19         | -7.914198 | 1.195047 | -6.62 | 0.000 | -10.66998 | -5.158414 |
| FE_C20         | -5.219565 | .6812369 | -7.66 | 0.000 | -6.7905   | -3.648629 |
| FE_C21         | 7.211811  | .7501515 | 9.61  | 0.000 | 5.481958  | 8.941663  |
| FE_C22         | 1.816441  | .5057793 | 3.59  | 0.007 | .6501119  | 2.98277   |
| FE_C23         | -.645257  | .3695764 | -1.75 | 0.119 | -1.497502 | .2069877  |
| FE_C24         | -1.673064 | .396247  | -4.22 | 0.003 | -2.586811 | -.7593166 |
| FE_C25         | 7.50589   | 1.136471 | 6.60  | 0.000 | 4.885182  | 10.1266   |
| FE_C26         | 3.372064  | .8269945 | 4.08  | 0.004 | 1.465011  | 5.279117  |
| FE_C27         | 7.162887  | .8488213 | 8.44  | 0.000 | 5.205501  | 9.120272  |
| FE_C28         | 11.35409  | 1.713997 | 6.62  | 0.000 | 7.401605  | 15.30657  |
| FE_C29         | .5641794  | .2666028 | 2.12  | 0.067 | -.0506078 | 1.178967  |
| FE_C30         | 10.52211  | 1.833086 | 5.74  | 0.000 | 6.295008  | 14.74922  |
| FE_C31         | 7.885879  | 1.299318 | 6.07  | 0.000 | 4.889646  | 10.88211  |
| FE_C32         | -6.372873 | .7891477 | -8.08 | 0.000 | -8.192651 | -4.553095 |
| FE_C33         | 1.808608  | .240029  | 7.53  | 0.000 | 1.2551    | 2.362116  |
| FE_C34         | 3.769788  | .7613989 | 4.95  | 0.001 | 2.013999  | 5.525577  |
| FE_C35         | 5.636799  | .6506937 | 8.66  | 0.000 | 4.136297  | 7.137302  |
| FE_C36         | .1416217  | .329203  | 0.43  | 0.678 | -.6175218 | .9007652  |
| FE_C37         | 3.869487  | .4934772 | 7.84  | 0.000 | 2.731526  | 5.007447  |
| FE_C38         | 6.578426  | .9765993 | 6.74  | 0.000 | 4.326384  | 8.830468  |
| FE_C39         | 2.00809   | .2017107 | 9.96  | 0.000 | 1.542944  | 2.473236  |
| FE_C40         | -5.391987 | .8476756 | -6.36 | 0.000 | -7.346731 | -3.437244 |
| FE_C41         | 4.482698  | .6501308 | 6.90  | 0.000 | 2.983494  | 5.981902  |
| FE_C42         | 5.970113  | 1.105278 | 5.40  | 0.001 | 3.421338  | 8.518888  |
| FE_C43         | -.0549383 | .2693748 | -0.20 | 0.843 | -.6761176 | .566241   |
| FE_C44         | 3.633027  | .419061  | 8.67  | 0.000 | 2.66667   | 4.599383  |
| FE_C45         | 4.135871  | .4708867 | 8.78  | 0.000 | 3.050004  | 5.221737  |
| FE_C46         | -4.797167 | .8011388 | -5.99 | 0.000 | -6.644597 | -2.949738 |
| FE_C47         | 7.109978  | 1.241205 | 5.73  | 0.000 | 4.247753  | 9.972203  |
| FE_C48         | -3.254173 | .5441851 | -5.98 | 0.000 | -4.509066 | -1.999279 |
| FE_C49         | -3.470142 | .8011553 | -4.33 | 0.003 | -5.31761  | -1.622675 |
| FE_C50         | 3.780726  | .3594639 | 10.52 | 0.000 | 2.951801  | 4.609651  |
| FE_C51         | -5.664967 | 1.134849 | -4.99 | 0.001 | -8.281934 | -3.048    |
| FE_C52         | .8617473  | .7096328 | 1.21  | 0.259 | -.7746688 | 2.498163  |
| FE_Y2          | -.0439803 | .0063539 | -6.92 | 0.000 | -.0586324 | -.0293283 |
| FE_Y3          | .0059342  | .0129927 | 0.46  | 0.660 | -.0240269 | .0358953  |
| FE_Y4          | -.0053588 | .0180575 | -0.30 | 0.774 | -.0469995 | .0362819  |
| FE_Y5          | -.0509172 | .0065919 | -7.72 | 0.000 | -.066118  | -.0357163 |
| FE_Y6          | .1076383  | .0571638 | 1.88  | 0.096 | -.0241816 | .2394583  |
| FE_Y7          | -.052365  | .0614923 | -0.85 | 0.419 | -.1941666 | .0894365  |
| FE_Y8          | -.0828074 | .0635282 | -1.30 | 0.229 | -.2293036 | .0636888  |
| FE_Y9          | -.1160865 | .0662712 | -1.75 | 0.118 | -.2689082 | .0367352  |
| _cons          | 4.548908  | .3146628 | 14.46 | 0.000 | 3.823295  | 5.274522  |

113 . estimates store M3

114 .

115 . \* Model IV

116 . xtsccl log\_app did c\_log\_GDPpc c\_log\_Pop15\_64 c\_YouthUnempl FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight], pooled lag(1)

|                                                |                  |   |         |
|------------------------------------------------|------------------|---|---------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468     |
| Method: <b>Pooled OLS</b>                      | Number of groups | = | 52      |
| Group variable (i): <b>DMC</b>                 | F( 63, 8)        | = | 3632.21 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000  |
|                                                | R-squared        | = | 0.9692  |
|                                                | Root MSE         | = | 0.2228  |

Drisc/Kraay

| log_app        | Coefficient | std. err. | t      | P> t  | [95% conf. interval] |           |
|----------------|-------------|-----------|--------|-------|----------------------|-----------|
| did            | -.960499    | .091648   | -10.48 | 0.000 | -1.17184             | -.7491584 |
| c_log_GDPpc    | -.6977425   | .2823517  | -2.47  | 0.039 | -1.348847            | -.0466382 |
| c_log_Pop15_64 | 3.212046    | .5204098  | 6.17   | 0.000 | 2.011979             | 4.412113  |
| c_YouthUnempl  | -.0140499   | .0044844  | -3.13  | 0.014 | -.0243908            | -.003709  |
| FE_C2          | 3.459467    | .6065053  | 5.70   | 0.000 | 2.060864             | 4.858071  |
| FE_C3          | 3.364003    | .4983211  | 6.75   | 0.000 | 2.214872             | 4.513134  |
| FE_C4          | 5.938747    | 1.288391  | 4.61   | 0.002 | 2.967712             | 8.909782  |
| FE_C5          | 12.43095    | 2.060336  | 6.03   | 0.000 | 7.679804             | 17.18209  |
| FE_C6          | 4.749261    | .586384   | 8.10   | 0.000 | 3.397057             | 6.101465  |
| FE_C7          | -10.28046   | 2.104597  | -4.88  | 0.001 | -15.13366            | -5.427248 |
| FE_C8          | 4.975758    | .9220879  | 5.40   | 0.001 | 2.849419             | 7.102096  |
| FE_C9          | 10.77686    | 1.5072    | 7.15   | 0.000 | 7.301253             | 14.25247  |
| FE_C10         | 3.126006    | .4132416  | 7.56   | 0.000 | 2.173069             | 4.078943  |
| FE_C11         | 4.626701    | .771406   | 6.00   | 0.000 | 2.847836             | 6.405567  |
| FE_C12         | -5.044478   | .93808    | -5.38  | 0.001 | -7.207695            | -2.881262 |
| FE_C13         | 8.698776    | 1.455514  | 5.98   | 0.000 | 5.342354             | 12.0552   |
| FE_C14         | 5.071243    | .7114114  | 7.13   | 0.000 | 3.430726             | 6.711761  |
| FE_C15         | -.6431554   | .4148564  | -1.55  | 0.160 | -1.599816            | .3135053  |
| FE_C16         | -1.9777     | .5661714  | -3.49  | 0.008 | -3.283294            | -.6721062 |
| FE_C17         | 4.212064    | .5637155  | 7.47   | 0.000 | 2.912134             | 5.511995  |
| FE_C18         | 3.152805    | .5090892  | 6.19   | 0.000 | 1.978843             | 4.326767  |
| FE_C19         | -8.56354    | 1.407716  | -6.08  | 0.000 | -11.80974            | -5.317341 |
| FE_C20         | -5.513535   | .7806175  | -7.06  | 0.000 | -7.313642            | -3.713428 |
| FE_C21         | 7.597739    | .891945   | 8.52   | 0.000 | 5.540911             | 9.654568  |
| FE_C22         | 1.84789     | .5506359  | 3.36   | 0.010 | .5781218             | 3.117659  |
| FE_C23         | -.5555282   | .3844795  | -1.44  | 0.186 | -1.44214             | .3310831  |
| FE_C24         | -1.85457    | .4569312  | -4.06  | 0.004 | -2.908255            | -.8008849 |
| FE_C25         | 7.783761    | 1.271715  | 6.12   | 0.000 | 4.851181             | 10.71634  |
| FE_C26         | 3.393707    | .8675834  | 3.91   | 0.004 | 1.393056             | 5.394358  |
| FE_C27         | 7.420479    | .9450032  | 7.85   | 0.000 | 5.241298             | 9.59966   |
| FE_C28         | 12.14765    | 2.009512  | 6.05   | 0.000 | 7.513708             | 16.78159  |
| FE_C29         | .3726432    | .3127199  | 1.19   | 0.268 | -.3484901            | 1.093776  |
| FE_C30         | 11.01429    | 2.063892  | 5.34   | 0.001 | 6.254949             | 15.77364  |
| FE_C31         | 8.23565     | 1.468281  | 5.61   | 0.001 | 4.849789             | 11.62151  |
| FE_C32         | -6.894344   | .949758   | -7.26  | 0.000 | -9.08449             | -4.704198 |
| FE_C33         | 1.850154    | .2704174  | 6.84   | 0.000 | 1.22657              | 2.473737  |
| FE_C34         | 3.955034    | .8547635  | 4.63   | 0.002 | 1.983946             | 5.926122  |
| FE_C35         | 5.954493    | .7643928  | 7.79   | 0.000 | 4.1918               | 7.717186  |
| FE_C36         | -.0052421   | .3625914  | -0.01  | 0.989 | -.8413795            | .8308952  |
| FE_C37         | 4.045075    | .5323692  | 7.60   | 0.000 | 2.817429             | 5.27272   |
| FE_C38         | 6.866066    | 1.110743  | 6.18   | 0.000 | 4.304689             | 9.427444  |
| FE_C39         | 2.049116    | .2010745  | 10.19  | 0.000 | 1.585438             | 2.512795  |
| FE_C40         | -5.68906    | .9704408  | -5.86  | 0.000 | -7.926901            | -3.45122  |
| FE_C41         | 4.686368    | .7444077  | 6.30   | 0.000 | 2.969761             | 6.402975  |
| FE_C42         | 6.243522    | 1.241174  | 5.03   | 0.001 | 3.38137              | 9.105674  |
| FE_C43         | .1156126    | .2635055  | 0.44   | 0.672 | -.4920321            | .7232573  |
| FE_C44         | 3.907012    | .5189021  | 7.53   | 0.000 | 2.710422             | 5.103603  |
| FE_C45         | 4.321221    | .5534602  | 7.81   | 0.000 | 3.04494              | 5.597503  |
| FE_C46         | -5.682004   | .98705    | -5.76  | 0.000 | -7.958145            | -3.405862 |
| FE_C47         | 7.34182     | 1.377999  | 5.33   | 0.001 | 4.164149             | 10.51949  |
| FE_C48         | -3.443386   | .6034402  | -5.71  | 0.000 | -4.834922            | -2.051851 |
| FE_C49         | -4.341169   | .9348953  | -4.64  | 0.002 | -6.497041            | -2.185296 |
| FE_C50         | 3.886191    | .4098855  | 9.48   | 0.000 | 2.940993             | 4.831388  |
| FE_C51         | -5.997047   | 1.277069  | -4.70  | 0.002 | -8.941974            | -3.052121 |
| FE_C52         | .3998798    | .7159798  | 0.56   | 0.592 | -1.251173            | 2.050932  |
| FE_Y2          | -.0563179   | .0082581  | -6.82  | 0.000 | -.0753611            | -.0372747 |
| FE_Y3          | -.0190169   | .0168004  | -1.13  | 0.290 | -.0577588            | .019725   |
| FE_Y4          | -.03918     | .023344   | -1.68  | 0.132 | -.0930113            | .0146512  |
| FE_Y5          | -.0648334   | .0100862  | -6.43  | 0.000 | -.0880923            | -.0415745 |
| FE_Y6          | .0889758    | .052536   | 1.69   | 0.129 | -.0321724            | .2101239  |
| FE_Y7          | -.0870014   | .056603   | -1.54  | 0.163 | -.2175282            | .0435255  |
| FE_Y8          | -.1251326   | .0590223  | -2.12  | 0.067 | -.2612382            | .0109731  |
| FE_Y9          | -.153105    | .06255    | -2.45  | 0.040 | -.2973455            | -.0088645 |
| _cons          | 4.523249    | .3392148  | 13.33  | 0.000 | 3.741018             | 5.30548   |

117 . estimates store M4

118 .

119 . \* Model V

120 . xtsccl log\_app did c\_log\_GDPpc c\_log\_Pop15\_64 c\_YouthUnempl did\_GDP did\_Pop did\_YU FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight], pooled  
> lag(1)

|                                                |                  |   |        |
|------------------------------------------------|------------------|---|--------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468    |
| Method: Pooled OLS                             | Number of groups | = | 52     |
| Group variable (i): DMC                        | F( 66, 8)        | = | 478.74 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000 |

R-squared = 0.9742  
Root MSE = 0.2049

| log_app        | Drisc/Kraay |          | t      | P> t  | [95% conf. interval] |           |
|----------------|-------------|----------|--------|-------|----------------------|-----------|
| Coefficient    | std. err.   |          |        |       |                      |           |
| did            | -1.20976    | .1194687 | -10.13 | 0.000 | -1.485256            | -.934265  |
| c_log_GDPpc    | -.003736    | .296827  | -0.01  | 0.990 | -.6882203            | .6807483  |
| c_log_Pop15_64 | 2.629088    | .4624549 | 5.69   | 0.000 | 1.562665             | 3.695511  |
| c_YouthUnempl  | -.0082929   | .0040356 | -2.05  | 0.074 | -.0175991            | .0010133  |
| did_GDP        | .808111     | .1365441 | 5.92   | 0.000 | .4932397             | 1.122982  |
| did_Pop        | .1000528    | .0130335 | 7.68   | 0.000 | .0699976             | .130108   |
| did_YU         | .0107662    | .0060521 | 1.78   | 0.113 | -.00319              | .0247223  |
| FE_C2          | 2.754771    | .5194243 | 5.30   | 0.001 | 1.556977             | 3.952566  |
| FE_C3          | 2.75553     | .4083768 | 6.75   | 0.000 | 1.813812             | 3.697249  |
| FE_C4          | 5.196133    | 1.129549 | 4.60   | 0.002 | 2.591388             | 7.800878  |
| FE_C5          | 9.861425    | 1.84225  | 5.35   | 0.001 | 5.613188             | 14.10966  |
| FE_C6          | 4.610685    | .5472641 | 8.42   | 0.000 | 3.348691             | 5.872678  |
| FE_C7          | -7.151949   | 1.930951 | -3.70  | 0.006 | -11.60473            | -2.699168 |
| FE_C8          | 4.27322     | .8119183 | 5.26   | 0.001 | 2.400933             | 6.145507  |
| FE_C9          | 9.293781    | 1.367803 | 6.79   | 0.000 | 6.139622             | 12.44794  |
| FE_C10         | 2.846737    | .3673867 | 7.75   | 0.000 | 1.999542             | 3.693933  |
| FE_C11         | 3.596245    | .6543748 | 5.50   | 0.001 | 2.087254             | 5.105236  |
| FE_C12         | -3.362304   | .8550501 | -3.93  | 0.004 | -5.334053            | -1.390555 |
| FE_C13         | 7.333196    | 1.324784 | 5.54   | 0.001 | 4.278239             | 10.38815  |
| FE_C14         | 4.13478     | .6108155 | 6.77   | 0.000 | 2.726237             | 5.543323  |
| FE_C15         | -.1944144   | .4188292 | -0.46  | 0.655 | -1.160236            | .7714075  |
| FE_C16         | -1.435811   | .555798  | -2.58  | 0.032 | -2.717483            | -.1541381 |
| FE_C17         | 3.99697     | .5284177 | 7.56   | 0.000 | 2.778437             | 5.215503  |
| FE_C18         | 3.001524    | .4665429 | 6.43   | 0.000 | 1.925674             | 4.077374  |
| FE_C19         | -6.120694   | 1.295109 | -4.73  | 0.001 | -9.107221            | -3.134168 |
| FE_C20         | -3.940668   | .7248342 | -5.44  | 0.001 | -5.612139            | -2.269197 |
| FE_C21         | 6.250001    | .7835485 | 7.98   | 0.000 | 4.443135             | 8.056867  |
| FE_C22         | 1.411147    | .4945113 | 2.85   | 0.021 | .2708017             | 2.551492  |
| FE_C23         | -.1883815   | .4092868 | -0.46  | 0.658 | -1.132198            | .7554354  |
| FE_C24         | -1.272116   | .4259648 | -2.99  | 0.017 | -2.254393            | -.2898397 |
| FE_C25         | 6.787112    | 1.167939 | 5.81   | 0.000 | 4.093841             | 9.480383  |
| FE_C26         | 3.485384    | .801939  | 4.35   | 0.002 | 1.636109             | 5.334659  |
| FE_C27         | 6.441467    | .8080738 | 7.97   | 0.000 | 4.578046             | 8.304889  |
| FE_C28         | 9.180595    | 1.745779 | 5.26   | 0.001 | 5.15482              | 13.20637  |
| FE_C29         | 1.011937    | .2833807 | 3.57   | 0.007 | .3584595             | 1.665414  |
| FE_C30         | 8.917516    | 1.885972 | 4.73   | 0.001 | 4.568457             | 13.26657  |
| FE_C31         | 7.072352    | 1.297347 | 5.45   | 0.001 | 4.080664             | 10.06404  |
| FE_C32         | -5.229331   | .8695243 | -6.01  | 0.000 | -7.234457            | -3.224204 |
| FE_C33         | 1.439688    | .1931848 | 7.45   | 0.000 | .9942029             | 1.885173  |
| FE_C34         | 3.143485    | .763106  | 4.12   | 0.003 | 1.383759             | 4.90321   |
| FE_C35         | 4.736368    | .6667194 | 7.10   | 0.000 | 3.19891              | 6.273826  |
| FE_C36         | .533119     | .3796179 | 1.40   | 0.198 | -.3422814            | 1.408519  |
| FE_C37         | 3.777694    | .4613908 | 8.19   | 0.000 | 2.713725             | 4.841663  |
| FE_C38         | 5.286612    | .9955593 | 5.31   | 0.001 | 2.990848             | 7.582376  |
| FE_C39         | 2.205141    | .1611785 | 13.68  | 0.000 | 1.833463             | 2.576819  |
| FE_C40         | -4.377846   | .8809612 | -4.97  | 0.001 | -6.409347            | -2.346346 |
| FE_C41         | 4.177481    | .6523646 | 6.40   | 0.000 | 2.673126             | 5.681836  |
| FE_C42         | 5.094248    | 1.139097 | 4.47   | 0.002 | 2.467485             | 7.721011  |
| FE_C43         | .4206941    | .2921344 | 1.44   | 0.188 | -.2529691            | 1.094357  |
| FE_C44         | 3.161034    | .4181702 | 7.56   | 0.000 | 2.196731             | 4.125336  |
| FE_C45         | 3.478489    | .4827478 | 7.21   | 0.000 | 2.365271             | 4.591708  |
| FE_C46         | -3.281255   | .9533532 | -3.44  | 0.009 | -5.479692            | -1.082819 |
| FE_C47         | 6.097847    | 1.228422 | 4.96   | 0.001 | 3.265101             | 8.930593  |
| FE_C48         | -2.337684   | .5697158 | -4.10  | 0.003 | -3.651451            | -1.023917 |
| FE_C49         | -1.971524   | .9473837 | -2.08  | 0.071 | -4.156195            | .2131469  |
| FE_C50         | 3.336497    | .3635183 | 9.18   | 0.000 | 2.498223             | 4.174772  |
| FE_C51         | -4.615568   | 1.147705 | -4.02  | 0.004 | -7.26218             | -1.968955 |
| FE_C52         | 1.68968     | .7577052 | 2.23   | 0.056 | -.0575911            | 3.436952  |
| FE_Y2          | -.0642947   | .0078924 | -8.15  | 0.000 | -.0824946            | -.0460948 |
| FE_Y3          | -.0355731   | .0160845 | -2.21  | 0.058 | -.0726639            | .0015177  |
| FE_Y4          | -.0618882   | .0223128 | -2.77  | 0.024 | -.1133415            | -.0104348 |
| FE_Y5          | -.0647604   | .0088786 | -7.29  | 0.000 | -.0852345            | -.0442862 |
| FE_Y6          | .0789673    | .0569566 | 1.39   | 0.203 | -.0523747            | .2103093  |
| FE_Y7          | -.107286    | .0533766 | -2.01  | 0.079 | -.2303726            | .0158006  |
| FE_Y8          | -.1489507   | .0549946 | -2.71  | 0.027 | -.2757685            | -.022133  |
| FE_Y9          | -.192099    | .0591816 | -3.25  | 0.012 | -.328572             | -.0556261 |
| _cons          | 4.667981    | .2929092 | 15.94  | 0.000 | 3.992532             | 5.343431  |

121 . estimates store M5

122 .



```

123 . esttab M2 M3 M4 M5, ///
> keep(did c_log_GDPpc c_log_Pop15_64 c_YouthUnempl did_GDP did_Pop did_YU) ///
> b(3) se(3) ///
> star(+ 0.10 * 0.05 ** 0.01 *** 0.001) ///
> stats(N r2, fmt(0 3)) ///
> mtitles("II" "III" "IV" "V")

```

|               | (1)<br>II            | (2)<br>III           | (3)<br>IV            | (4)<br>V             |
|---------------|----------------------|----------------------|----------------------|----------------------|
| did           | -1.049***<br>(0.111) | -0.980***<br>(0.101) | -0.960***<br>(0.092) | -1.210***<br>(0.119) |
| c_log_GDPpc   | -0.727*<br>(0.256)   | -0.459<br>(0.259)    | -0.698*<br>(0.282)   | -0.004<br>(0.297)    |
| c_log_Pop~64  |                      | 3.075***<br>(0.460)  | 3.212***<br>(0.520)  | 2.629***<br>(0.462)  |
| c_YouthUnempl |                      |                      | -0.014*<br>(0.004)   | -0.008+<br>(0.004)   |
| did_GDP       |                      |                      |                      | 0.808***<br>(0.137)  |
| did_Pop       |                      |                      |                      | 0.100***<br>(0.013)  |
| did_YU        |                      |                      |                      | 0.011<br>(0.006)     |
| N             | 468                  | 468                  | 468                  | 468                  |
| r2            | 0.963                | 0.968                | 0.969                | 0.974                |

Standard errors in parentheses

+ p<0.10, \* p<0.05, \*\* p<0.01, \*\*\* p<0.001

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124 .
125 . *****
126 . * 8. Saturated country-specific DiD variables
127 . * Use ISO3 codes for readable and stable variable names
128 . *****
129 .
130 . capture drop SAT_*

131 .
132 . levelsof iso3c if treated_factor != "CONTROL", local(eu_iso)
    `AUT' `BEL' `BGR' `CYP' `CZE' `DEU' `DNK' `ESP' `EST' `FIN' `FRA' `GRC' `HRV' `HUN' `ITA' `LTU' `LUX' `L
    > VA' `MLT' `NLD' `POL' `PRT' `ROU' `SVK' `SVN' `SWE'

133 .
134 . foreach c of local eu_iso {
1.   gen double SAT_`c' = Post*(iso3c == "`c'")
2.   quietly levelsof Domicile_named_country if iso3c == "`c'", local(country_name) clean
3.   label variable SAT_`c' "`country_name' x Post"
4.   }

135 .
136 . *****
137 . * 9. Saturated models S1-S4
138 . *****
139 .
140 . local SATVARS

141 . foreach c of local eu_iso {
1.   local SATVARS `SATVARS' SAT_`c'
2.   }

142 .
143 . * S1 saturated DiD
144 . xtsccl log_app `SATVARS' FE_C2-FE_C52 FE_Y2-FE_Y9 [aw=regression_weight], pooled lag(1)

```

|                                                |                  |   |         |
|------------------------------------------------|------------------|---|---------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468     |
| Method: Pooled OLS                             | Number of groups | = | 52      |
| Group variable (i): DMC                        | F( 85, 8)        | = | 3872.85 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000  |
|                                                | R-squared        | = | 0.9838  |
|                                                | Root MSE         | = | 0.1665  |

| log_app | Drisc/Kraay |           | t      | P> t  | [95% conf. interval] |           |
|---------|-------------|-----------|--------|-------|----------------------|-----------|
|         | Coefficient | std. err. |        |       |                      |           |
| SAT_AUT | -.5292352   | .0589537  | -8.98  | 0.000 | -.6651828            | -.3932876 |
| SAT_BEL | -.5808624   | .0737412  | -7.88  | 0.000 | -.75091              | -.4108148 |
| SAT_BGR | -1.855027   | .1506795  | -12.31 | 0.000 | -2.202495            | -1.50756  |
| SAT_CYP | -.8871006   | .070117   | -12.65 | 0.000 | -1.048791            | -.7254106 |
| SAT_CZE | -1.121458   | .0647607  | -17.32 | 0.000 | -1.270796            | -.9721196 |
| SAT_DEU | -.6589969   | .0782589  | -8.42  | 0.000 | -.8394623            | -.4785314 |
| SAT_DNK | -1.034555   | .1328164  | -7.79  | 0.000 | -1.340831            | -.7282804 |
| SAT_ESP | -.4919003   | .0952959  | -5.16  | 0.001 | -.7116531            | -.2721476 |
| SAT_EST | -1.411935   | .133106   | -10.61 | 0.000 | -1.718878            | -1.104993 |
| SAT_FIN | -1.680281   | .2196102  | -7.65  | 0.000 | -2.186703            | -1.173859 |
| SAT_FRA | -.5749273   | .0694052  | -8.28  | 0.000 | -.7349761            | -.4148785 |
| SAT_GRC | -.8187998   | .0672905  | -12.17 | 0.000 | -.973972             | -.6636277 |
| SAT_HRV | -1.615906   | .1282872  | -12.60 | 0.000 | -1.911737            | -1.320075 |
| SAT_HUN | -1.075652   | .0864086  | -12.45 | 0.000 | -1.27491             | -.876393  |
| SAT_ITA | -.8861136   | .1196062  | -7.41  | 0.000 | -1.161926            | -.6103013 |
| SAT_LTU | -2.163162   | .3506138  | -6.17  | 0.000 | -2.971679            | -1.354645 |
| SAT_LUX | -.5053025   | .0565495  | -8.94  | 0.000 | -.6357058            | -.3748992 |
| SAT_LVA | -1.47435    | .1552769  | -9.49  | 0.000 | -1.832419            | -1.11628  |
| SAT_MLT | -.350701    | .0444564  | -7.89  | 0.000 | -.4532177            | -.2481844 |
| SAT_NLD | -.5205676   | .023599   | -22.06 | 0.000 | -.5749869            | -.4661484 |
| SAT_POL | -1.612018   | .1667213  | -9.67  | 0.000 | -1.996479            | -1.227558 |
| SAT_PRT | -.8628774   | .4251858  | -2.03  | 0.077 | -1.843358            | .1176028  |
| SAT_ROU | -1.652883   | .1551301  | -10.65 | 0.000 | -2.010613            | -1.295152 |
| SAT_SVK | -1.47216    | .1184047  | -12.43 | 0.000 | -1.745201            | -1.199118 |
| SAT_SVN | -1.201293   | .1223242  | -9.82  | 0.000 | -1.483373            | -.9192127 |
| SAT_SWE | -.9102995   | .1445131  | -6.30  | 0.000 | -1.243547            | -.5770518 |
| FE_C2   | -.0975137   | .0673612  | -1.45  | 0.186 | -.2528489            | .0578216  |
| FE_C3   | .4835526    | .0662184  | 7.30   | 0.000 | .3308528             | .6362524  |
| FE_C4   | -1.371569   | .0331581  | -41.36 | 0.000 | -1.448032            | -1.295107 |
| FE_C5   | -.5953534   | .0840681  | -7.08  | 0.000 | -.7892149            | -.4014919 |
| FE_C6   | 1.195527    | .0829677  | 14.41  | 0.000 | 1.004204             | 1.386851  |
| FE_C7   | 3.555301    | .1183995  | 30.03  | 0.000 | 3.282272             | 3.828331  |
| FE_C8   | -.6026252   | .0613339  | -9.83  | 0.000 | -.7440615            | -.461189  |
| FE_C9   | 1.471952    | .0873135  | 16.86  | 0.000 | 1.270607             | 1.673297  |
| FE_C10  | .5510391    | .0659543  | 8.35   | 0.000 | .3989481             | .7031301  |
| FE_C11  | -.2479885   | .0710444  | -3.49  | 0.008 | -.4118171            | -.0841598 |
| FE_C12  | .2334908    | .0790045  | 2.96   | 0.018 | .0513061             | .4156754  |
| FE_C13  | -.4749375   | .0695767  | -6.83  | 0.000 | -.6353817            | -.3144933 |
| FE_C14  | .254021     | .0923324  | 2.75   | 0.025 | .0411021             | .4669399  |
| FE_C15  | 2.090806    | .0594871  | 35.15  | 0.000 | 1.953629             | 2.227983  |
| FE_C16  | 1.648434    | .0685499  | 24.05  | 0.000 | 1.490358             | 1.806511  |
| FE_C17  | 1.285816    | .0610616  | 21.06  | 0.000 | 1.145008             | 1.426624  |
| FE_C18  | .3974523    | .0659984  | 6.02   | 0.000 | .2452598             | .5496449  |
| FE_C19  | .1933797    | .0408929  | 4.73   | 0.001 | .0990805             | .2876789  |
| FE_C20  | -.6459632   | .0462036  | -13.98 | 0.000 | -.7525089            | -.5394176 |
| FE_C21  | 1.979359    | .0399064  | 49.60  | 0.000 | 1.887334             | 2.071383  |
| FE_C22  | -1.446572   | .0386917  | -37.39 | 0.000 | -1.535795            | -1.357348 |
| FE_C23  | 1.916439    | .0707077  | 27.10  | 0.000 | 1.753387             | 2.079491  |
| FE_C24  | .8572076    | .0921189  | 9.31   | 0.000 | .6447809             | 1.069634  |
| FE_C25  | -.1019542   | .0605852  | -1.68  | 0.131 | -.2416639            | .0377555  |
| FE_C26  | -.5213461   | .1032967  | -5.05  | 0.001 | -.7595486            | -.2831436 |
| FE_C27  | 1.076487    | .0749473  | 14.36  | 0.000 | .9036585             | 1.249316  |
| FE_C28  | -.4290141   | .0648516  | -6.62  | 0.000 | -.5785622            | -.279466  |
| FE_C29  | 1.907597    | .0885357  | 21.55  | 0.000 | 1.703433             | 2.111176  |
| FE_C30  | -1.658828   | .0646979  | -25.64 | 0.000 | -1.808021            | -1.509634 |
| FE_C31  | -.6609445   | .1016487  | -6.50  | 0.000 | -.8953468            | -.4265422 |
| FE_C32  | -.9315127   | .0542647  | -17.17 | 0.000 | -1.056647            | -.8063782 |
| FE_C33  | .3583993    | .0602654  | 5.95   | 0.000 | .219427              | .4973716  |
| FE_C34  | -1.142046   | .0424933  | -26.88 | 0.000 | -1.240036            | -1.044057 |
| FE_C35  | .64298      | .1865497  | 3.45   | 0.009 | .2127957             | 1.073164  |
| FE_C36  | 1.847731    | .0726356  | 25.44  | 0.000 | 1.680233             | 2.015229  |
| FE_C37  | 1.166986    | .1356976  | 8.60   | 0.000 | .8540666             | 1.479905  |
| FE_C38  | .0622113    | .0157552  | 3.95   | 0.004 | .0258798             | .0985428  |
| FE_C39  | 1.663121    | .0607785  | 27.36  | 0.000 | 1.522966             | 1.803277  |
| FE_C40  | .2270981    | .1209402  | 1.88   | 0.097 | -.0517904            | .5059866  |
| FE_C41  | .2610943    | .0983574  | 2.65   | 0.029 | .0342817             | .4879069  |
| FE_C42  | -1.513271   | .0895603  | -16.90 | 0.000 | -1.719798            | -1.306745 |
| FE_C43  | 1.786952    | .0799625  | 22.35  | 0.000 | 1.602558             | 1.971346  |
| FE_C44  | .652123     | .0716765  | 9.10   | 0.000 | .4868366             | .8174094  |
| FE_C45  | .6946695    | .0727531  | 9.55   | 0.000 | .5269005             | .8624386  |
| FE_C46  | -1.430745   | .0567992  | -25.19 | 0.000 | -1.561724            | -1.299766 |
| FE_C47  | -1.37235    | .135625   | -10.12 | 0.000 | -1.685102            | -1.059599 |
| FE_C48  | .8084768    | .1216971  | 6.64   | 0.000 | .5278427             | 1.089111  |
| FE_C49  | -1.02789    | .0605746  | -16.97 | 0.000 | -1.167575            | -.8882045 |
| FE_C50  | 1.412938    | .0924911  | 15.28  | 0.000 | 1.199653             | 1.626223  |
| FE_C51  | 2.148921    | .0512713  | 41.91  | 0.000 | 2.030689             | 2.267152  |

|        |           |          |          |       |           |           |
|--------|-----------|----------|----------|-------|-----------|-----------|
| FE_C52 | -.1079512 | .055373  | -1.95    | 0.087 | -.2356416 | .0197392  |
| FE_Y2  | -.0469583 | 3.81e-13 | -1.2e+11 | 0.000 | -.0469583 | -.0469583 |
| FE_Y3  | -.0007601 | 5.35e-13 | -1.4e+09 | 0.000 | -.0007601 | -.0007601 |
| FE_Y4  | -.0116871 | 5.22e-13 | -2.2e+10 | 0.000 | -.0116871 | -.0116871 |
| FE_Y5  | -.0280463 | 5.35e-13 | -5.2e+10 | 0.000 | -.0280463 | -.0280463 |
| FE_Y6  | .146377   | .0577558 | 2.53     | 0.035 | .0131918  | .2795622  |
| FE_Y7  | -.0132739 | .0577558 | -0.23    | 0.824 | -.1464591 | .1199113  |
| FE_Y8  | -.0318182 | .0577558 | -0.55    | 0.597 | -.1650035 | .101367   |
| FE_Y9  | -.0556913 | .0577558 | -0.96    | 0.363 | -.1888766 | .0774939  |
| _cons  | 6.523507  | .0587576 | 111.02   | 0.000 | 6.388012  | 6.659003  |

145 . estimates store S1

146 .

147 . \* S2 plus GDPpc

148 . xtsccl log\_app `SATVARS' c\_log\_GDPpc FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight], pooled lag(1)

|                                                |                  |   |         |
|------------------------------------------------|------------------|---|---------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468     |
| Method: Pooled OLS                             | Number of groups | = | 52      |
| Group variable (i): DMC                        | F( 86, 8)        | = | 9973.15 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000  |
|                                                | R-squared        | = | 0.9839  |
|                                                | Root MSE         | = | 0.1663  |

| log_app     | Drisc/Kraay |           |        |       |           | [95% conf. interval] |
|-------------|-------------|-----------|--------|-------|-----------|----------------------|
|             | Coefficient | std. err. | t      | P> t  |           |                      |
| SAT_AUT     | -.5214528   | .0598004  | -8.72  | 0.000 | -.6593528 | -.3835528            |
| SAT_BEL     | -.5825905   | .0709451  | -8.21  | 0.000 | -.7461901 | -.418991             |
| SAT_BGR     | -1.887609   | .1393018  | -13.55 | 0.000 | -2.20884  | -1.566379            |
| SAT_CYP     | -.9204714   | .0593963  | -15.50 | 0.000 | -1.057439 | -.7835034            |
| SAT_CZE     | -1.120939   | .0604454  | -18.54 | 0.000 | -1.260327 | -.9815517            |
| SAT_DEU     | -.6500607   | .0798979  | -8.14  | 0.000 | -.8343055 | -.4658158            |
| SAT_DNK     | -1.037175   | .1285393  | -8.07  | 0.000 | -1.333587 | -.7407624            |
| SAT_ESP     | -.4876474   | .0974152  | -5.01  | 0.001 | -.7122872 | -.2630075            |
| SAT_EST     | -1.412059   | .1252798  | -11.27 | 0.000 | -1.700955 | -1.123163            |
| SAT_FIN     | -1.669078   | .2224158  | -7.50  | 0.000 | -2.18197  | -1.156187            |
| SAT_FRA     | -.5712674   | .0691252  | -8.26  | 0.000 | -.7306704 | -.4118644            |
| SAT_GRC     | -.8351015   | .0600485  | -13.91 | 0.000 | -.9735736 | -.6966295            |
| SAT_HRV     | -1.656679   | .118406   | -13.99 | 0.000 | -1.929724 | -1.383634            |
| SAT_HUN     | -1.097764   | .0831272  | -13.21 | 0.000 | -1.289456 | -.906072             |
| SAT_ITA     | -.8920014   | .1139629  | -7.83  | 0.000 | -1.1548   | -.6292025            |
| SAT_LTU     | -2.187302   | .3376788  | -6.48  | 0.000 | -2.965991 | -1.408613            |
| SAT_LUX     | -.4898083   | .0570846  | -8.58  | 0.000 | -.6214456 | -.358171             |
| SAT_LVA     | -1.487602   | .1469142  | -10.13 | 0.000 | -1.826387 | -1.148818            |
| SAT_MLT     | -.3759914   | .0501796  | -7.49  | 0.000 | -.4917057 | -.2602771            |
| SAT_NLD     | -.5228356   | .0233519  | -22.39 | 0.000 | -.5766851 | -.4689861            |
| SAT_POL     | -1.649463   | .1539959  | -10.71 | 0.000 | -2.004578 | -1.294348            |
| SAT_PRT     | -.8672682   | .4247653  | -2.04  | 0.075 | -1.846779 | .1122423             |
| SAT_ROU     | -1.679655   | .1465599  | -11.46 | 0.000 | -2.017623 | -1.341687            |
| SAT_SVK     | -1.48231    | .1144944  | -12.95 | 0.000 | -1.746334 | -1.218285            |
| SAT SVN     | -1.217604   | .1180738  | -10.31 | 0.000 | -1.489883 | -.9453253            |
| SAT_SWE     | -.9058813   | .1434372  | -6.32  | 0.000 | -1.236648 | -.5751147            |
| c_log_GDPpc | .2794038    | .256467   | 1.09   | 0.308 | -.3120101 | .8708176             |
| FE_C2       | -.1234499   | .0805328  | -1.53  | 0.164 | -.3091588 | .0622591             |
| FE_C3       | .4752529    | .0705377  | 6.74   | 0.000 | .3125928  | .6379131             |
| FE_C4       | -1.039227   | .2942557  | -3.53  | 0.008 | -1.717782 | -.3606725            |
| FE_C5       | -.6780806   | .1316134  | -5.15  | 0.001 | -.9815817 | -.3745795            |
| FE_C6       | 1.399042    | .1567317  | 8.93   | 0.000 | 1.037618  | 1.760466             |
| FE_C7       | 3.861385    | .3619035  | 10.67  | 0.000 | 3.026834  | 4.695936             |
| FE_C8       | -.448869    | .1375371  | -3.26  | 0.011 | -.7660301 | -.1317079            |
| FE_C9       | 1.553218    | .080418   | 19.31  | 0.000 | 1.367774  | 1.738663             |
| FE_C10      | .6142044    | .0853101  | 7.20   | 0.000 | .4174791  | .8109298             |
| FE_C11      | -.2859446   | .091477   | -3.13  | 0.014 | -.496891  | -.0749982            |
| FE_C12      | .6077914    | .3409204  | 1.78   | 0.112 | -.1783725 | 1.393955             |
| FE_C13      | -.377157    | .0802199  | -4.70  | 0.002 | -.5621444 | -.1921697            |
| FE_C14      | .2607754    | .0904113  | 2.88   | 0.020 | .0522867  | .4692641             |
| FE_C15      | 2.118463    | .0572032  | 37.03  | 0.000 | 1.986552  | 2.250374             |
| FE_C16      | 1.626568    | .080322   | 20.25  | 0.000 | 1.441345  | 1.811791             |
| FE_C17      | 1.446917    | .1369676  | 10.56  | 0.000 | 1.13107   | 1.762765             |
| FE_C18      | .5328129    | .137636   | 3.87   | 0.005 | .2154237  | .8502022             |
| FE_C19      | .6188065    | .4117023  | 1.50   | 0.171 | -.3305808 | 1.568194             |
| FE_C20      | -.2848064   | .3171823  | -0.90  | 0.395 | -1.01623  | .4466173             |
| FE_C21      | 1.820988    | .1726618  | 10.55  | 0.000 | 1.422829  | 2.219146             |
| FE_C22      | -1.373677   | .0692333  | -19.84 | 0.000 | -1.533329 | -1.214024            |
| FE_C23      | 1.96294     | .0607917  | 32.29  | 0.000 | 1.822754  | 2.103126             |
| FE_C24      | .8975277    | .0935793  | 9.59   | 0.000 | .6817334  | 1.113322             |

|        |           |          |        |       |           |           |
|--------|-----------|----------|--------|-------|-----------|-----------|
| FE_C25 | .0444387  | .135751  | 0.33   | 0.752 | -.2686037 | .3574811  |
| FE_C26 | -.1070841 | .4220493 | -0.25  | 0.806 | -1.080332 | .8661632  |
| FE_C27 | 1.172283  | .092399  | 12.69  | 0.000 | .9592101  | 1.385355  |
| FE_C28 | -.6637375 | .2454    | -2.70  | 0.027 | -1.229631 | -.0978442 |
| FE_C29 | 2.086118  | .1359468 | 15.35  | 0.000 | 1.772624  | 2.399612  |
| FE_C30 | -1.628352 | .0695663 | -23.41 | 0.000 | -1.788772 | -1.467932 |
| FE_C31 | -.421302  | .1692038 | -2.49  | 0.038 | -.8114867 | -.0311173 |
| FE_C32 | -.6511682 | .2883775 | -2.26  | 0.054 | -1.316168 | .0138316  |
| FE_C33 | .3179571  | .0811712 | 3.92   | 0.004 | .130776   | .5051383  |
| FE_C34 | -1.085008 | .0576123 | -18.83 | 0.000 | -1.217862 | -.952154  |
| FE_C35 | .5174191  | .235707  | 2.20   | 0.059 | -.0261222 | 1.06096   |
| FE_C36 | 1.977251  | .1422979 | 13.90  | 0.000 | 1.649112  | 2.305391  |
| FE_C37 | 1.279316  | .1984099 | 6.45   | 0.000 | .8217822  | 1.73685   |
| FE_C38 | -.124287  | .1574873 | -0.79  | 0.453 | -.4874534 | .2388794  |
| FE_C39 | 1.80878   | .1282864 | 14.10  | 0.000 | 1.512951  | 2.104609  |
| FE_C40 | .3496587  | .1488297 | 2.35   | 0.047 | .0064568  | .6928605  |
| FE_C41 | .3911803  | .1655714 | 2.36   | 0.046 | .0093721  | .7729886  |
| FE_C42 | -1.43015  | .0872651 | -16.39 | 0.000 | -1.631384 | -1.228917 |
| FE_C43 | 1.85384   | .1123087 | 16.51  | 0.000 | 1.594856  | 2.112824  |
| FE_C44 | .6378739  | .0797733 | 8.00   | 0.000 | .4539164  | .8218314  |
| FE_C45 | .5989225  | .1274161 | 4.70   | 0.002 | .3051005  | .8927445  |
| FE_C46 | -.6417632 | .6889511 | -0.93  | 0.379 | -2.230487 | .9469609  |
| FE_C47 | -1.208305 | .1377861 | -8.77  | 0.000 | -1.526041 | -.89057   |
| FE_C48 | .9878653  | .2118649 | 4.66   | 0.002 | .4993039  | 1.476427  |
| FE_C49 | -.1671945 | .8062855 | -0.21  | 0.841 | -2.026492 | 1.692103  |
| FE_C50 | 1.35846   | .0861995 | 15.76  | 0.000 | 1.159684  | 1.557236  |
| FE_C51 | 2.098736  | .028608  | 73.36  | 0.000 | 2.032766  | 2.164706  |
| FE_C52 | .5758003  | .6730836 | 0.86   | 0.417 | -.9763334 | 2.127934  |
| FE_Y2  | -.0537179 | .0062047 | -8.66  | 0.000 | -.068026  | -.0394098 |
| FE_Y3  | -.0145974 | .0127014 | -1.15  | 0.284 | -.0438868 | .0146919  |
| FE_Y4  | -.0308411 | .0175816 | -1.75  | 0.117 | -.0713844 | .0097021  |
| FE_Y5  | -.0331997 | .0047303 | -7.02  | 0.000 | -.0441078 | -.0222916 |
| FE_Y6  | .1312724  | .0661167 | 1.99   | 0.082 | -.0211929 | .2837378  |
| FE_Y7  | -.0367756 | .0717903 | -0.51  | 0.622 | -.2023243 | .128773   |
| FE_Y8  | -.0588743 | .0743021 | -0.79  | 0.451 | -.2302153 | .1124667  |
| FE_Y9  | -.0871227 | .0774712 | -1.12  | 0.293 | -.2657715 | .0915261  |
| _cons  | 6.413388  | .0990284 | 64.76  | 0.000 | 6.185029  | 6.641748  |

149 . estimates store S2

150 .

151 . \* S3 plus GDPpc and population

152 . xtsccl log\_app `SATVARS' c\_log\_GDPpc c\_log\_Pop15\_64 FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight], pooled lag(1)

|                                                |                  |   |          |
|------------------------------------------------|------------------|---|----------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468      |
| Method: Pooled OLS                             | Number of groups | = | 52       |
| Group variable (i): DMC                        | F( 87, 8)        | = | 75411.34 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000   |
|                                                | R-squared        | = | 0.9850   |
|                                                | Root MSE         | = | 0.1606   |

| log_app | Drisc/Kraay |           |        |       |                      |           |
|---------|-------------|-----------|--------|-------|----------------------|-----------|
|         | Coefficient | std. err. | t      | P> t  | [95% conf. interval] |           |
| SAT_AUT | -.4988132   | .052307   | -9.54  | 0.000 | -.6194334            | -.378193  |
| SAT_BEL | -.5702803   | .0647846  | -8.80  | 0.000 | -.7196738            | -.4208868 |
| SAT_BGR | -1.770381   | .0963179  | -18.38 | 0.000 | -1.992491            | -1.548271 |
| SAT_CYP | -.9475419   | .0610642  | -15.52 | 0.000 | -1.088356            | -.8067276 |
| SAT_CZE | -1.06958    | .0716909  | -14.92 | 0.000 | -1.234899            | -.9042603 |
| SAT_DEU | -.5805322   | .0553622  | -10.49 | 0.000 | -.7081975            | -.4528668 |
| SAT_DNK | -1.025301   | .1236787  | -8.29  | 0.000 | -1.310504            | -.7400969 |
| SAT_ESP | -.4900811   | .1000824  | -4.90  | 0.001 | -.7208715            | -.2592907 |
| SAT_EST | -1.392716   | .1215611  | -11.46 | 0.000 | -1.673036            | -1.112395 |
| SAT_FIN | -1.624849   | .2077049  | -7.82  | 0.000 | -2.103818            | -1.145881 |
| SAT_FRA | -.5409222   | .05838    | -9.27  | 0.000 | -.6755467            | -.4062976 |
| SAT_GRC | -.7353715   | .0356456  | -20.63 | 0.000 | -.8175704            | -.6531726 |
| SAT_HRV | -1.528228   | .0978612  | -15.62 | 0.000 | -1.753896            | -1.30256  |
| SAT_HUN | -1.007874   | .0804329  | -12.53 | 0.000 | -1.193352            | -.8223952 |
| SAT_ITA | -.8239817   | .0923631  | -8.92  | 0.000 | -1.036971            | -.610992  |
| SAT_LTU | -2.157345   | .3343084  | -6.45  | 0.000 | -2.928261            | -1.386428 |
| SAT_LUX | -.5751222   | .0862595  | -6.67  | 0.000 | -.7740368            | -.3762075 |
| SAT_LVA | -1.382666   | .1272535  | -10.87 | 0.000 | -1.676113            | -1.089219 |
| SAT_MLT | -.5227425   | .0646648  | -8.08  | 0.000 | -.6718598            | -.3736252 |
| SAT_NLD | -.5217203   | .0241412  | -21.61 | 0.000 | -.5773899            | -.4660506 |
| SAT_POL | -1.51574    | .1330021  | -11.40 | 0.000 | -1.822444            | -1.209037 |
| SAT_PRT | -.8332499   | .4217186  | -1.98  | 0.084 | -1.805735            | .139235   |
| SAT_ROU | -1.577122   | .1255943  | -12.56 | 0.000 | -1.866743            | -1.287501 |

|                |           |          |        |       |           |           |
|----------------|-----------|----------|--------|-------|-----------|-----------|
| SAT_SVK        | -1.384913 | .1168486 | -11.85 | 0.000 | -1.654366 | -1.115459 |
| SAT_SVN        | -1.170733 | .0997521 | -11.74 | 0.000 | -1.400762 | -.9407044 |
| SAT_SWE        | -.9164239 | .1427438 | -6.42  | 0.000 | -1.245592 | -.5872562 |
| c_log_GDPpc    | .3323243  | .2651158 | 1.25   | 0.245 | -.2790338 | .9436823  |
| c_log_Pop15_64 | 1.645876  | .3623824 | 4.54   | 0.002 | .810221   | 2.481532  |
| FE_C2          | 1.562953  | .3353233 | 4.66   | 0.002 | .7896956  | 2.336209  |
| FE_C3          | 1.803916  | .2580784 | 6.99   | 0.000 | 1.208786  | 2.399046  |
| FE_C4          | 2.99118   | .935041  | 3.20   | 0.013 | .834972   | 5.147389  |
| FE_C5          | 5.791264  | 1.379985 | 4.20   | 0.003 | 2.609012  | 8.973517  |
| FE_C6          | 3.636183  | .5116547 | 7.11   | 0.000 | 2.456305  | 4.816061  |
| FE_C7          | -2.787745 | 1.422301 | -1.96  | 0.086 | -6.067577 | .4920873  |
| FE_C8          | 2.625941  | .6842464 | 3.84   | 0.005 | 1.048066  | 4.203816  |
| FE_C9          | 6.35427   | 1.028936 | 6.18   | 0.000 | 3.981538  | 8.727001  |
| FE_C10         | 2.063723  | .2899525 | 7.12   | 0.000 | 1.395091  | 2.732355  |
| FE_C11         | 2.165057  | .5110239 | 4.24   | 0.003 | .9866334  | 3.34348   |
| FE_C12         | -1.627201 | .630312  | -2.58  | 0.033 | -3.080703 | -.1736995 |
| FE_C13         | 4.529512  | 1.045696 | 4.33   | 0.003 | 2.118132  | 6.940892  |
| FE_C14         | 2.841703  | .5396505 | 5.27   | 0.001 | 1.597267  | 4.086139  |
| FE_C15         | .5938832  | .375438  | 1.58   | 0.152 | -.2718784 | 1.459645  |
| FE_C16         | -.3245188 | .4736852 | -0.69  | 0.513 | -1.416839 | .7678013  |
| FE_C17         | 2.924144  | .3405707 | 8.59   | 0.000 | 2.138787  | 3.709502  |
| FE_C18         | 2.107451  | .356333  | 5.91   | 0.000 | 1.285746  | 2.929156  |
| FE_C19         | -3.266585 | .9138127 | -3.57  | 0.007 | -5.373841 | -1.159329 |
| FE_C20         | -2.347076 | .5024419 | -4.67  | 0.002 | -3.505709 | -1.188443 |
| FE_C21         | 4.452748  | .5428989 | 8.20   | 0.000 | 3.200821  | 5.704675  |
| FE_C22         | .4510288  | .4262864 | 1.06   | 0.321 | -.5319895 | 1.434047  |
| FE_C23         | .5859286  | .3433014 | 1.71   | 0.126 | -.2057258 | 1.377583  |
| FE_C24         | -.4138882 | .341119  | -1.21  | 0.260 | -1.20051  | .3727337  |
| FE_C25         | 4.338288  | .9288016 | 4.67   | 0.002 | 2.196467  | 6.480108  |
| FE_C26         | 2.420032  | .7540632 | 3.21   | 0.012 | .6811596  | 4.158905  |
| FE_C27         | 4.789606  | .7922056 | 6.05   | 0.000 | 2.962776  | 6.616435  |
| FE_C28         | 5.325456  | 1.280565 | 4.16   | 0.003 | 2.372467  | 8.278446  |
| FE_C29         | 1.557802  | .2082623 | 7.48   | 0.000 | 1.077548  | 2.038056  |
| FE_C30         | 4.828472  | 1.386031 | 3.48   | 0.008 | 1.632278  | 8.024666  |
| FE_C31         | 4.41023   | 1.062346 | 4.15   | 0.003 | 1.960455  | 6.860004  |
| FE_C32         | -3.264365 | .5896838 | -5.54  | 0.001 | -4.624179 | -1.904552 |
| FE_C33         | .9401597  | .1062787 | 8.85   | 0.000 | .6950805  | 1.185239  |
| FE_C34         | 1.605403  | .6129143 | 2.62   | 0.031 | .19202    | 3.018786  |
| FE_C35         | 3.056509  | .4377416 | 6.98   | 0.000 | 2.047075  | 4.065943  |
| FE_C36         | 1.275525  | .2480887 | 5.14   | 0.001 | .703431   | 1.847618  |
| FE_C37         | 2.801038  | .3931141 | 7.13   | 0.000 | 1.894516  | 3.707561  |
| FE_C38         | 3.164635  | .7311208 | 4.33   | 0.003 | 1.478668  | 4.850603  |
| FE_C39         | 2.250957  | .1621415 | 13.88  | 0.000 | 1.877058  | 2.624856  |
| FE_C40         | -2.52736  | .6963912 | -3.63  | 0.007 | -4.133241 | -.9214791 |
| FE_C41         | 2.859157  | .5377909 | 5.32   | 0.001 | 1.619009  | 4.099305  |
| FE_C42         | 2.688818  | .8966621 | 3.00   | 0.017 | .6211114  | 4.756524  |
| FE_C43         | .8262933  | .2864883 | 2.88   | 0.020 | .1656501  | 1.486937  |
| FE_C44         | 2.208584  | .3092892 | 7.14   | 0.000 | 1.495362  | 2.921806  |
| FE_C45         | 2.338728  | .3359739 | 6.96   | 0.000 | 1.563971  | 3.113485  |
| FE_C46         | -1.600426 | .7054504 | -2.27  | 0.053 | -3.227198 | .0263451  |
| FE_C47         | 3.507882  | 1.0074   | 3.48   | 0.008 | 1.184814  | 5.83095   |
| FE_C48         | -.9951999 | .3768867 | -2.64  | 0.030 | -1.864302 | -.1260976 |
| FE_C49         | -.5544591 | .7855709 | -0.71  | 0.500 | -2.365989 | 1.257071  |
| FE_C50         | 2.567758  | .3015586 | 8.51   | 0.000 | 1.872363  | 3.263153  |
| FE_C51         | -2.137893 | .9141246 | -2.34  | 0.048 | -4.245868 | -.0299177 |
| FE_C52         | 1.825878  | .7584779 | 2.41   | 0.043 | .0768253  | 3.574932  |
| FE_Y2          | -.0593502 | .0066595 | -8.91  | 0.000 | -.074707  | -.0439934 |
| FE_Y3          | -.0258067 | .0136097 | -1.90  | 0.095 | -.0571907 | .0055773  |
| FE_Y4          | -.0479302 | .0189522 | -2.53  | 0.035 | -.091634  | -.0042264 |
| FE_Y5          | -.0509523 | .0066551 | -7.66  | 0.000 | -.066299  | -.0356056 |
| FE_Y6          | .0943869  | .0596937 | 1.58   | 0.152 | -.043267  | .2320409  |
| FE_Y7          | -.082827  | .0652764 | -1.27  | 0.240 | -.2333546 | .0677006  |
| FE_Y8          | -.1150947 | .0675518 | -1.70  | 0.127 | -.2708693 | .04068    |
| FE_Y9          | -.153056  | .0706909 | -2.17  | 0.062 | -.3160694 | .0099574  |
| _cons          | 5.23861   | .2535365 | 20.66  | 0.000 | 4.653953  | 5.823266  |

153 . estimates store S3

154 .

155 . \* S4 plus GDPpc, population, and youth unemployment

156 . xtsccl log\_app `SATVARS' c\_log\_GDPpc c\_log\_Pop15\_64 c\_YouthUnempl FE\_C2-FC52 FE\_Y2-FCY9 [aw=regression\_weight], pooled lag(1)

|                                                |                  |   |         |
|------------------------------------------------|------------------|---|---------|
| Regression with Driscoll-Kraay standard errors | Number of obs    | = | 468     |
| Method: Pooled OLS                             | Number of groups | = | 52      |
| Group variable (i): DMC                        | F( 88, 8)        | = | 2619.63 |
| maximum lag: 1                                 | Prob > F         | = | 0.0000  |
|                                                | R-squared        | = | 0.9850  |
|                                                | Root MSE         | = | 0.1608  |

| log_app        | Drisc/Kraay |          | t      | P> t  | [95% conf. interval] |           |
|----------------|-------------|----------|--------|-------|----------------------|-----------|
| Coefficient    | std. err.   |          |        |       |                      |           |
| SAT_AUT        | -.5041833   | .0577919 | -8.72  | 0.000 | -.6374516            | -.3709151 |
| SAT_BEL        | -.5751065   | .0688235 | -8.36  | 0.000 | -.7338137            | -.4163993 |
| SAT_BGR        | -1.780743   | .1060235 | -16.80 | 0.000 | -2.025234            | -1.536252 |
| SAT_CYP        | -.9436781   | .054403  | -17.35 | 0.000 | -1.069132            | -.8182245 |
| SAT_CZE        | -1.07543    | .0727041 | -14.79 | 0.000 | -1.243086            | -.9077735 |
| SAT_DEU        | -.5835169   | .0590054 | -9.89  | 0.000 | -.7195837            | -.4474501 |
| SAT_DNK        | -1.029637   | .1285107 | -8.01  | 0.000 | -1.325983            | -.7332904 |
| SAT_ESP        | -.4764577   | .1032979 | -4.61  | 0.002 | -.7146629            | -.2382524 |
| SAT_EST        | -1.406561   | .1331062 | -10.57 | 0.000 | -1.713505            | -1.099618 |
| SAT_FIN        | -1.622217   | .208437  | -7.78  | 0.000 | -2.102874            | -1.141561 |
| SAT_FRA        | -.5372455   | .0577163 | -9.31  | 0.000 | -.6703395            | -.4041516 |
| SAT_GRC        | -.7194327   | .0420664 | -17.10 | 0.000 | -.816438             | -.6224273 |
| SAT_HRV        | -1.529496   | .101721  | -15.04 | 0.000 | -1.764065            | -1.294926 |
| SAT_HUN        | -1.02015    | .0790884 | -12.90 | 0.000 | -1.202529            | -.8377722 |
| SAT_ITA        | -.8110422   | .0846775 | -9.58  | 0.000 | -1.006309            | -.6157755 |
| SAT_LTU        | -2.164376   | .342453  | -6.32  | 0.000 | -2.954074            | -1.374678 |
| SAT_LUX        | -.5771991   | .0911056 | -6.34  | 0.000 | -.7872891            | -.3671091 |
| SAT_LVA        | -1.38907    | .1304505 | -10.65 | 0.000 | -1.68989             | -1.088251 |
| SAT_MLT        | -.5258676   | .0656011 | -8.02  | 0.000 | -.6771439            | -.3745913 |
| SAT_NLD        | -.526531    | .0274798 | -19.16 | 0.000 | -.5898994            | -.4631625 |
| SAT_POL        | -1.523933   | .1360431 | -11.20 | 0.000 | -1.837649            | -1.210217 |
| SAT_PRT        | -.8357402   | .4252306 | -1.97  | 0.085 | -1.816324            | .1448433  |
| SAT_ROU        | -1.596503   | .1368689 | -11.66 | 0.000 | -1.912123            | -1.280882 |
| SAT_SVK        | -1.396035   | .111203  | -12.55 | 0.000 | -1.65247             | -1.139601 |
| SAT_SVN        | -1.177089   | .1079757 | -10.90 | 0.000 | -1.426081            | -.9280965 |
| SAT_SWE        | -.9274465   | .1530419 | -6.06  | 0.000 | -1.280362            | -.5745313 |
| c_log_GDPpc    | .375761     | .2387925 | 1.57   | 0.154 | -.1748956            | .9264176  |
| c_log_Pop15_64 | 1.617236    | .3435501 | 4.71   | 0.002 | .8250077             | 2.409463  |
| c_YouthUnempl  | .0022337    | .002854  | 0.78   | 0.456 | -.0043476            | .008815   |
| FE_C2          | 1.533824    | .3105235 | 4.94   | 0.001 | .8177557             | 2.249893  |
| FE_C3          | 1.768358    | .2328328 | 7.59   | 0.000 | 1.231445             | 2.305271  |
| FE_C4          | 2.910615    | .9060709 | 3.21   | 0.012 | .8212121             | 5.000019  |
| FE_C5          | 5.639522    | 1.28051  | 4.40   | 0.002 | 2.686661             | 8.592383  |
| FE_C6          | 3.62689     | .501481  | 7.23   | 0.000 | 2.470472             | 4.783307  |
| FE_C7          | -2.626308   | 1.297603 | -2.02  | 0.078 | -5.618587            | .3659705  |
| FE_C8          | 2.570263    | .6606923 | 3.89   | 0.005 | 1.046703             | 4.093822  |
| FE_C9          | 6.261221    | .9705609 | 6.45   | 0.000 | 4.023103             | 8.499338  |
| FE_C10         | 2.057942    | .2818199 | 7.30   | 0.000 | 1.408064             | 2.70782   |
| FE_C11         | 2.117736    | .4737949 | 4.47   | 0.002 | 1.025163             | 3.210309  |
| FE_C12         | -1.553824   | .5696058 | -2.73  | 0.026 | -2.867337            | -.2403102 |
| FE_C13         | 4.455785    | .9973678 | 4.47   | 0.002 | 2.155851             | 6.75572   |
| FE_C14         | 2.781953    | .499331  | 5.57   | 0.001 | 1.630494             | 3.933412  |
| FE_C15         | .6029021    | .3652732 | 1.65   | 0.137 | -.2394194            | 1.445224  |
| FE_C16         | -.282435    | .4544395 | -0.62  | 0.552 | -1.330374            | .7655043  |
| FE_C17         | 2.861759    | .3410011 | 8.39   | 0.000 | 2.075409             | 3.648109  |
| FE_C18         | 2.102286    | .348347  | 6.04   | 0.000 | 1.298997             | 2.905576  |
| FE_C19         | -3.138643   | .8082364 | -3.88  | 0.005 | -5.00244             | -1.274847 |
| FE_C20         | -2.284455   | .4442247 | -5.14  | 0.001 | -3.308839            | -1.260071 |
| FE_C21         | 4.377312    | .4815425 | 9.09   | 0.000 | 3.266873             | 5.487751  |
| FE_C22         | .440006     | .4133168 | 1.06   | 0.318 | -.5131043            | 1.393116  |
| FE_C23         | .5709749    | .3452795 | 1.65   | 0.137 | -.225241             | 1.367191  |
| FE_C24         | -.3788059   | .3221635 | -1.18  | 0.273 | -1.121716            | .3641045  |
| FE_C25         | 4.280575    | .8946721 | 4.78   | 0.001 | 2.217458             | 6.343693  |
| FE_C26         | 2.414662    | .7471732 | 3.23   | 0.012 | .6916779             | 4.137647  |
| FE_C27         | 4.737185    | .7573789 | 6.25   | 0.000 | 2.990666             | 6.483704  |
| FE_C28         | 5.169663    | 1.16847  | 4.42   | 0.002 | 2.475167             | 7.864159  |
| FE_C29         | 1.594082    | .1903729 | 8.37   | 0.000 | 1.155082             | 2.033083  |
| FE_C30         | 4.724302    | 1.313909 | 3.60   | 0.007 | 1.694422             | 7.754182  |
| FE_C31         | 4.339629    | 1.02215  | 4.25   | 0.003 | 1.982546             | 6.696712  |
| FE_C32         | -3.164962   | .5046619 | -6.27  | 0.000 | -4.328714            | -2.001209 |
| FE_C33         | .9308979    | .0953401 | 9.76   | 0.000 | .7110431             | 1.150753  |
| FE_C34         | 1.566039    | .5837855 | 2.68   | 0.028 | .2198275             | 2.912251  |
| FE_C35         | 2.992975    | .3780998 | 7.92   | 0.000 | 2.121076             | 3.864875  |
| FE_C36         | 1.306088    | .2214862 | 5.90   | 0.000 | .7953396             | 1.816836  |
| FE_C37         | 2.768777    | .3919395 | 7.06   | 0.000 | 1.864963             | 3.672592  |
| FE_C38         | 3.101544    | .6780979 | 4.57   | 0.002 | 1.537847             | 4.66524   |
| FE_C39         | 2.252395    | .159445  | 14.13  | 0.000 | 1.884714             | 2.620076  |
| FE_C40         | -2.465774   | .6546805 | -3.77  | 0.005 | -3.97547             | -.956078  |
| FE_C41         | 2.822566    | .524959  | 5.38   | 0.001 | 1.612009             | 4.033124  |
| FE_C42         | 2.631297    | .8554186 | 3.08   | 0.015 | .6586982             | 4.603896  |
| FE_C43         | .7973856    | .3063158 | 2.60   | 0.031 | .0910201             | 1.503751  |
| FE_C44         | 2.161841    | .2772205 | 7.80   | 0.000 | 1.522569             | 2.801113  |
| FE_C45         | 2.300136    | .3010316 | 7.64   | 0.000 | 1.605956             | 2.994316  |
| FE_C46         | -1.439658   | .5899464 | -2.44  | 0.041 | -2.800077            | -.0792395 |
| FE_C47         | 3.454958    | .9677553 | 3.57   | 0.007 | 1.22331              | 5.686605  |

|        |           |          |       |       |           |           |
|--------|-----------|----------|-------|-------|-----------|-----------|
| FE_C48 | -.9532801 | .3319356 | -2.87 | 0.021 | -1.718725 | -.1878352 |
| FE_C49 | -.3967721 | .6786998 | -0.58 | 0.575 | -1.961857 | 1.168312  |
| FE_C50 | 2.544896  | .2883842 | 8.82  | 0.000 | 1.87988   | 3.209911  |
| FE_C51 | -2.068677 | .8672227 | -2.39 | 0.044 | -4.068496 | -.0688582 |
| FE_C52 | 1.908152  | .7169843 | 2.66  | 0.029 | .2547835  | 3.561521  |
| FE_Y2  | -.0575039 | .0080767 | -7.12 | 0.000 | -.0761287 | -.0388791 |
| FE_Y3  | -.022077  | .0164778 | -1.34 | 0.217 | -.0600748 | .0159209  |
| FE_Y4  | -.0428749 | .0227733 | -1.88 | 0.097 | -.0953901 | .0096404  |
| FE_Y5  | -.0487722 | .0077261 | -6.31 | 0.000 | -.0665886 | -.0309559 |
| FE_Y6  | .0971967  | .0621702 | 1.56  | 0.157 | -.0461679 | .2405614  |
| FE_Y7  | -.0776118 | .0696015 | -1.12 | 0.297 | -.2381132 | .0828897  |
| FE_Y8  | -.1086879 | .0727965 | -1.49 | 0.174 | -.2765568 | .059181   |
| FE_Y9  | -.1475424 | .0752409 | -1.96 | 0.086 | -.3210481 | .0259633  |
| _cons  | 5.245273  | .2477392 | 21.17 | 0.000 | 4.673986  | 5.816561  |

```
157 . estimates store S4
```

```
158 .
```

```
159 . esttab S1 S2 S3 S4, ///
```

```
> keep(`SATVARS' c_log_GDPpc c_log_Pop15_64 c_YouthUnempl) ///  
> label ///  
> b(3) se(3) ///  
> star(+ 0.10 * 0.05 ** 0.01 *** 0.001) ///  
> stats(N r2, fmt(0 3)) ///  
> mtitles("S1 Saturated DiD" "S2 + GDPpc" "S3 + GDPpc + Pop" "S4 + GDPpc + Pop + YouthUnempl")
```

|                      | (1)<br>S1 Saturat~D  | (2)<br>S2 + GDPpc    | (3)<br>S3 + GDPpc~p  | (4)<br>S4 + GDPpc~l  |
|----------------------|----------------------|----------------------|----------------------|----------------------|
| Austria x Post       | -0.529***<br>(0.059) | -0.521***<br>(0.060) | -0.499***<br>(0.052) | -0.504***<br>(0.058) |
| Belgium x Post       | -0.581***<br>(0.074) | -0.583***<br>(0.071) | -0.570***<br>(0.065) | -0.575***<br>(0.069) |
| Bulgaria x Post      | -1.855***<br>(0.151) | -1.888***<br>(0.139) | -1.770***<br>(0.096) | -1.781***<br>(0.106) |
| Cyprus (European U~t | -0.887***<br>(0.070) | -0.920***<br>(0.059) | -0.948***<br>(0.061) | -0.944***<br>(0.054) |
| Czech Republic x P~t | -1.121***<br>(0.065) | -1.121***<br>(0.060) | -1.070***<br>(0.072) | -1.075***<br>(0.073) |
| Germany x Post       | -0.659***<br>(0.078) | -0.650***<br>(0.080) | -0.581***<br>(0.055) | -0.584***<br>(0.059) |
| Denmark x Post       | -1.035***<br>(0.133) | -1.037***<br>(0.129) | -1.025***<br>(0.124) | -1.030***<br>(0.129) |
| Spain x Post         | -0.492***<br>(0.095) | -0.488**<br>(0.097)  | -0.490**<br>(0.100)  | -0.476**<br>(0.103)  |
| Estonia x Post       | -1.412***<br>(0.133) | -1.412***<br>(0.125) | -1.393***<br>(0.122) | -1.407***<br>(0.133) |
| Finland x Post       | -1.680***<br>(0.220) | -1.669***<br>(0.222) | -1.625***<br>(0.208) | -1.622***<br>(0.208) |
| France x Post        | -0.575***<br>(0.069) | -0.571***<br>(0.069) | -0.541***<br>(0.058) | -0.537***<br>(0.058) |
| Greece x Post        | -0.819***<br>(0.067) | -0.835***<br>(0.060) | -0.735***<br>(0.036) | -0.719***<br>(0.042) |
| Croatia x Post       | -1.616***<br>(0.128) | -1.657***<br>(0.118) | -1.528***<br>(0.098) | -1.529***<br>(0.102) |
| Hungary x Post       | -1.076***<br>(0.086) | -1.098***<br>(0.083) | -1.008***<br>(0.080) | -1.020***<br>(0.079) |
| Italy x Post         | -0.886***<br>(0.120) | -0.892***<br>(0.114) | -0.824***<br>(0.092) | -0.811***<br>(0.085) |
| Lithuania x Post     | -2.163***<br>(0.351) | -2.187***<br>(0.338) | -2.157***<br>(0.334) | -2.164***<br>(0.342) |

|                    |                      |                      |                      |                      |
|--------------------|----------------------|----------------------|----------------------|----------------------|
| Luxembourg x Post  | -0.505***<br>(0.057) | -0.490***<br>(0.057) | -0.575***<br>(0.086) | -0.577***<br>(0.091) |
| Latvia x Post      | -1.474***<br>(0.155) | -1.488***<br>(0.147) | -1.383***<br>(0.127) | -1.389***<br>(0.130) |
| Malta x Post       | -0.351***<br>(0.044) | -0.376***<br>(0.050) | -0.523***<br>(0.065) | -0.526***<br>(0.066) |
| Netherlands x Post | -0.521***<br>(0.024) | -0.523***<br>(0.023) | -0.522***<br>(0.024) | -0.527***<br>(0.027) |
| Poland x Post      | -1.612***<br>(0.167) | -1.649***<br>(0.154) | -1.516***<br>(0.133) | -1.524***<br>(0.136) |
| Portugal x Post    | -0.863+<br>(0.425)   | -0.867+<br>(0.425)   | -0.833+<br>(0.422)   | -0.836+<br>(0.425)   |
| Romania x Post     | -1.653***<br>(0.155) | -1.680***<br>(0.147) | -1.577***<br>(0.126) | -1.597***<br>(0.137) |
| Slovakia x Post    | -1.472***<br>(0.118) | -1.482***<br>(0.114) | -1.385***<br>(0.117) | -1.396***<br>(0.111) |
| Slovenia x Post    | -1.201***<br>(0.122) | -1.218***<br>(0.118) | -1.171***<br>(0.100) | -1.177***<br>(0.108) |
| Sweden x Post      | -0.910***<br>(0.145) | -0.906***<br>(0.143) | -0.916***<br>(0.143) | -0.927***<br>(0.153) |
| c_log_GDPpc        |                      | 0.279<br>(0.256)     | 0.332<br>(0.265)     | 0.376<br>(0.239)     |
| c_log_Pop15_64     |                      |                      | 1.646**<br>(0.362)   | 1.617**<br>(0.344)   |
| c_YouthUnempl      |                      |                      |                      | 0.002<br>(0.003)     |
| N                  | 468                  | 468                  | 468                  | 468                  |
| r2                 | 0.984                | 0.984                | 0.985                | 0.985                |

Standard errors in parentheses

+ p&lt;0.10, \* p&lt;0.05, \*\* p&lt;0.01, \*\*\* p&lt;0.001

```

160 .
161 . *****
162 . * 10. Optional point-estimate verification against
163 . *      ordinary weighted LSDV for Model V and S4
164 . *****
165 .
166 . reg log_app did c_log_GDPpc c_log_Pop15_64 c_YouthUnempl did_GDP did_Pop did_YU FE_C2-FE_C52 FE_Y2-FE_Y9 [aw=regression_weight]
(sum of wgt is 468)

```

| Source   | SS         | df  | MS         | Number of obs | = | 468    |
|----------|------------|-----|------------|---------------|---|--------|
| Model    | 635.468188 | 66  | 9.62830588 | F(66, 401)    | = | 229.27 |
| Residual | 16.8404889 | 401 | .041996232 | Prob > F      | = | 0.0000 |
|          |            |     |            | R-squared     | = | 0.9742 |
|          |            |     |            | Adj R-squared | = | 0.9699 |
| Total    | 652.308677 | 467 | 1.39680659 | Root MSE      | = | .20493 |

| log_app        | Coefficient | Std. err. | t      | P> t  | [95% conf. interval] |           |
|----------------|-------------|-----------|--------|-------|----------------------|-----------|
| did            | -1.20976    | .0569067  | -21.26 | 0.000 | -1.321633            | -1.097888 |
| c_log_GDPpc    | -.003736    | .2465494  | -0.02  | 0.988 | -.4884267            | .4809547  |
| c_log_Pop15_64 | 2.629088    | .3741643  | 7.03   | 0.000 | 1.893519             | 3.364657  |
| c_YouthUnempl  | -.0082929   | .0041555  | -2.00  | 0.047 | -.0164623            | -.0001235 |
| did_GDP        | .808111     | .1063821  | 7.60   | 0.000 | .5989747             | 1.017247  |
| did_Pop        | .1000528    | .021295   | 4.70   | 0.000 | .0581889             | .1419166  |
| did_YU         | .0107662    | .0043661  | 2.47   | 0.014 | .0021829             | .0193495  |
| FE_C2          | 2.754771    | .4043047  | 6.81   | 0.000 | 1.95995              | 3.549593  |
| FE_C3          | 2.75553     | .3303707  | 8.34   | 0.000 | 2.106056             | 3.405005  |
| FE_C4          | 5.196133    | 669.2432  | 0.01   | 0.994 | -1310.467            | 1320.86   |
| FE_C5          | 9.861425    | 8376.646  | 0.00   | 0.999 | -16457.77            | 16477.49  |
| FE_C6          | 4.610685    | .5507727  | 8.37   | 0.000 | 3.527922             | 5.693447  |
| FE_C7          | -7.151949   | 1.548723  | -4.62  | 0.000 | -10.19658            | -4.10732  |
| FE_C8          | 4.27322     | .7287796  | 5.86   | 0.000 | 2.840514             | 5.705926  |
| FE_C9          | 9.293781    | 1.098568  | 8.46   | 0.000 | 7.13411              | 11.45345  |
| FE_C10         | 2.846737    | .355331   | 8.01   | 0.000 | 2.148193             | 3.545282  |



|        |           |           |       |       |           |           |
|--------|-----------|-----------|-------|-------|-----------|-----------|
| FE_C11 | 3.596245  | .5742225  | 6.26  | 0.000 | 2.467383  | 4.725108  |
| FE_C12 | -3.362304 | .6123166  | -5.49 | 0.000 | -4.566055 | -2.158552 |
| FE_C13 | 7.333196  | 1.124516  | 6.52  | 0.000 | 5.122514  | 9.543879  |
| FE_C14 | 4.13478   | .6056378  | 6.83  | 0.000 | 2.944158  | 5.325402  |
| FE_C15 | -.1944144 | .3710858  | -0.52 | 0.601 | -.923931  | .5351022  |
| FE_C16 | -1.435811 | .4636976  | -3.10 | 0.002 | -2.347393 | -.5242287 |
| FE_C17 | 3.99697   | .3955252  | 10.11 | 0.000 | 3.219408  | 4.774532  |
| FE_C18 | 3.001524  | .3956542  | 7.59  | 0.000 | 2.223709  | 3.77934   |
| FE_C19 | -6.120694 | .9731458  | -6.29 | 0.000 | -8.033799 | -4.20759  |
| FE_C20 | -3.940668 | .728.2087 | -0.01 | 0.996 | -1435.524 | 1427.643  |
| FE_C21 | 6.250001  | .646196   | 9.67  | 0.000 | 4.979646  | 7.520356  |
| FE_C22 | 1.411147  | .4357793  | 3.24  | 0.001 | .5544496  | 2.267844  |
| FE_C23 | -.1883815 | .3382701  | -0.56 | 0.578 | -.8533859 | .4766229  |
| FE_C24 | -1.272116 | .3217524  | -3.95 | 0.000 | -1.904648 | -.6395839 |
| FE_C25 | 6.787112  | .9953871  | 6.82  | 0.000 | 4.830283  | 8.743941  |
| FE_C26 | 3.485384  | .6916963  | 5.04  | 0.000 | 2.12558   | 4.845188  |
| FE_C27 | 6.441467  | .8337972  | 7.73  | 0.000 | 4.802307  | 8.080627  |
| FE_C28 | 9.180595  | 1.398998  | 6.56  | 0.000 | 6.430308  | 11.93088  |
| FE_C29 | 1.011937  | .2287978  | 4.42  | 0.000 | .5621436  | 1.46173   |
| FE_C30 | 8.917516  | 1.459097  | 6.11  | 0.000 | 6.049082  | 11.78595  |
| FE_C31 | 7.072352  | 1.122395  | 6.30  | 0.000 | 4.865838  | 9.278866  |
| FE_C32 | -5.229331 | .6666836  | -7.84 | 0.000 | -6.539962 | -3.918699 |
| FE_C33 | 1.439688  | .1913525  | 7.52  | 0.000 | 1.063508  | 1.815867  |
| FE_C34 | 3.143485  | .6248464  | 5.03  | 0.000 | 1.915101  | 4.371869  |
| FE_C35 | 4.736368  | .6108993  | 7.75  | 0.000 | 3.535403  | 5.937333  |
| FE_C36 | .533119   | .2275359  | 2.34  | 0.020 | .0858067  | .9804313  |
| FE_C37 | 3.777694  | .3798311  | 9.95  | 0.000 | 3.030985  | 4.524403  |
| FE_C38 | 5.286612  | .772744   | 6.84  | 0.000 | 3.767476  | 6.805747  |
| FE_C39 | 2.205141  | .2001876  | 11.02 | 0.000 | 1.811593  | 2.598689  |
| FE_C40 | -4.377846 | .6724244  | -6.51 | 0.000 | -5.699764 | -3.055929 |
| FE_C41 | 4.177481  | .5894563  | 7.09  | 0.000 | 3.01867   | 5.336292  |
| FE_C42 | 5.094248  | .9470803  | 5.38  | 0.000 | 3.232385  | 6.95611   |
| FE_C43 | .4206941  | .281652   | 1.49  | 0.136 | -.1330047 | .974393   |
| FE_C44 | 3.161034  | .3847313  | 8.22  | 0.000 | 2.404691  | 3.917376  |
| FE_C45 | 3.478489  | .102.4149 | 0.03  | 0.973 | -197.8587 | 204.8157  |
| FE_C46 | -3.281255 | .7614808  | -4.31 | 0.000 | -4.778248 | -1.784262 |
| FE_C47 | 6.097847  | 1.083934  | 5.63  | 0.000 | 3.966944  | 8.228751  |
| FE_C48 | -2.337684 | .201.8337 | -0.01 | 0.991 | -399.1221 | 394.4467  |
| FE_C49 | -1.971524 | .1319.823 | -0.00 | 0.999 | -2596.608 | 2592.665  |
| FE_C50 | 3.336497  | .3053892  | 10.93 | 0.000 | 2.736134  | 3.936861  |
| FE_C51 | -4.615568 | .9714693  | -4.75 | 0.000 | -6.525377 | -2.705759 |
| FE_C52 | 1.68968   | .6705724  | 2.52  | 0.012 | .3714036  | 3.007957  |
| FE_Y2  | -.0642947 | .0407432  | -1.58 | 0.115 | -.1443916 | .0158022  |
| FE_Y3  | -.0355731 | .0424405  | -0.84 | 0.402 | -.1190068 | .0478605  |
| FE_Y4  | -.0618882 | .0443886  | -1.39 | 0.164 | -.1491516 | .0253753  |
| FE_Y5  | -.0647604 | .0408089  | -1.59 | 0.113 | -.1449864 | .0154657  |
| FE_Y6  | .0789673  | .047329   | 1.67  | 0.096 | -.0140766 | .1720112  |
| FE_Y7  | -.107286  | .0506143  | -2.12 | 0.035 | -.2067885 | -.0077835 |
| FE_Y8  | -.1489507 | .0527861  | -2.82 | 0.005 | -.2527228 | -.0451786 |
| FE_Y9  | -.192099  | .0546958  | -3.51 | 0.000 | -.2996254 | -.0845726 |
| _cons  | 4.667981  | .3017814  | 15.47 | 0.000 | 4.07471   | 5.261253  |

167 . estimates store M5\_WLS\_check

168 .

169 . reg log\_app `SATVARS' c\_log\_GDPpc c\_log\_Pop15\_64 c\_YouthUnempl FE\_C2-FE\_C52 FE\_Y2-FE\_Y9 [aw=regression\_weight]  
(sum of wgt is 468)

| Source   | SS         | df  | MS         | Number of obs | = | 468    |
|----------|------------|-----|------------|---------------|---|--------|
| Model    | 642.512599 | 88  | 7.30127954 | F(88, 379)    | = | 282.48 |
| Residual | 9.79607793 | 379 | .025847171 | Prob > F      | = | 0.0000 |
|          |            |     |            | R-squared     | = | 0.9850 |
|          |            |     |            | Adj R-squared | = | 0.9815 |
| Total    | 652.308677 | 467 | 1.39680659 | Root MSE      | = | .16077 |

| log_app | Coefficient | Std. err. | t      | P> t  | [95% conf. interval] |           |
|---------|-------------|-----------|--------|-------|----------------------|-----------|
| SAT_AUT | -.5041833   | .110501   | -4.56  | 0.000 | -.7214552            | -.2869115 |
| SAT_BEL | -.5751065   | .1102423  | -5.22  | 0.000 | -.7918696            | -.3583433 |
| SAT_BGR | -1.780743   | .11575    | -15.38 | 0.000 | -2.008336            | -1.55315  |
| SAT_CYP | -.9436781   | .1126581  | -8.38  | 0.000 | -1.165191            | -.7221648 |
| SAT_CZE | -1.07543    | .1107799  | -9.71  | 0.000 | -1.29325             | -.8576093 |
| SAT_DEU | -.5835169   | .1109739  | -5.26  | 0.000 | -.8017186            | -.3653152 |
| SAT_DNK | -1.029637   | .1101897  | -9.34  | 0.000 | -1.246296            | -.8129767 |
| SAT_ESP | -.4764577   | .1123789  | -4.24  | 0.000 | -.6974219            | -.2554934 |
| SAT_EST | -1.406561   | .112477   | -12.51 | 0.000 | -1.627718            | -1.185404 |
| SAT_FIN | -1.622217   | .1105872  | -14.67 | 0.000 | -1.839659            | -1.404776 |
| SAT_FRA | -.5372455   | .1102599  | -4.87  | 0.000 | -.7540433            | -.3204477 |

|                |           |           |        |       |           |           |
|----------------|-----------|-----------|--------|-------|-----------|-----------|
| SAT_GRC        | -.7194327 | .1153367  | -6.24  | 0.000 | -.9462126 | -.4926527 |
| SAT_HRV        | -1.529496 | .1161068  | -13.17 | 0.000 | -1.75779  | -1.301201 |
| SAT_HUN        | -1.02015  | .1142097  | -8.93  | 0.000 | -1.244715 | -.7955863 |
| SAT_ITA        | -.8110422 | .1129079  | -7.18  | 0.000 | -1.033047 | -.5890379 |
| SAT_LTU        | -2.164376 | .1119785  | -19.33 | 0.000 | -2.384553 | -1.944199 |
| SAT_LUX        | -.5771991 | .1116549  | -5.17  | 0.000 | -.7967398 | -.3576584 |
| SAT_LVA        | -1.38907  | .1125833  | -12.34 | 0.000 | -1.610436 | -1.167704 |
| SAT_MLT        | -.5258676 | .1148218  | -4.58  | 0.000 | -.7516351 | -.3001001 |
| SAT_NLD        | -.526531  | .1102209  | -4.78  | 0.000 | -.743252  | -.3098099 |
| SAT_POL        | -1.523933 | .1165927  | -13.07 | 0.000 | -1.753183 | -1.294684 |
| SAT_PRT        | -.8357402 | .1102142  | -7.58  | 0.000 | -1.052448 | -.6190323 |
| SAT_ROU        | -1.596503 | .1178877  | -13.54 | 0.000 | -1.828298 | -1.364707 |
| SAT_SVK        | -1.396035 | .1132546  | -12.33 | 0.000 | -1.618722 | -1.173349 |
| SAT_SVN        | -1.177089 | .1113747  | -10.57 | 0.000 | -1.396079 | -.9580992 |
| SAT_SWE        | -.9274465 | .1115638  | -8.31  | 0.000 | -1.146808 | -.7080851 |
| c_log_GDPpc    | .375761   | .2090884  | 1.80   | 0.073 | -.0353576 | .7868796  |
| c_log_Pop15_64 | 1.617236  | .3139754  | 5.15   | 0.000 | .9998836  | 2.234587  |
| c_YouthUnempl  | .0022337  | .003815   | 0.59   | 0.559 | -.0052675 | .0097348  |
| FE_C2          | 1.533824  | .3400519  | 4.51   | 0.000 | .8651995  | 2.202449  |
| FE_C3          | 1.768358  | .2797402  | 6.32   | 0.000 | 1.218321  | 2.318395  |
| FE_C4          | 2.910615  | .525.0316 | 0.01   | 0.996 | -1029.429 | 1035.25   |
| FE_C5          | 5.639522  | .6571.606 | 0.00   | 0.999 | -12915.74 | 12927.01  |
| FE_C6          | 3.62689   | .4582933  | 7.91   | 0.000 | 2.725774  | 4.528006  |
| FE_C7          | -2.626308 | 1.306478  | -2.01  | 0.045 | -5.195162 | -.0574545 |
| FE_C8          | 2.570263  | .6067475  | 4.24   | 0.000 | 1.37725   | 3.763275  |
| FE_C9          | 6.261221  | .9267094  | 6.76   | 0.000 | 4.439085  | 8.083356  |
| FE_C10         | 2.057942  | .29749    | 6.92   | 0.000 | 1.473004  | 2.642879  |
| FE_C11         | 2.117736  | .4820733  | 4.39   | 0.000 | 1.169863  | 3.06561   |
| FE_C12         | -1.553824 | .5192532  | -2.99  | 0.003 | -2.574801 | -.5328457 |
| FE_C13         | 4.455785  | .9419341  | 4.73   | 0.000 | 2.603714  | 6.307857  |
| FE_C14         | 2.781953  | .5087338  | 5.47   | 0.000 | 1.781659  | 3.782247  |
| FE_C15         | .6029021  | .3082083  | 1.96   | 0.051 | -.0031102 | 1.208915  |
| FE_C16         | -.282435  | .3904076  | -0.72  | 0.470 | -1.050071 | .4852012  |
| FE_C17         | 2.861759  | .3366881  | 8.50   | 0.000 | 2.199748  | 3.52377   |
| FE_C18         | 2.102286  | .330021   | 6.37   | 0.000 | 1.453385  | 2.751188  |
| FE_C19         | -3.138643 | .8260446  | -3.80  | 0.000 | -4.762848 | -1.514439 |
| FE_C20         | -2.284455 | .571.2908 | -0.00  | 0.997 | -1125.581 | 1121.012  |
| FE_C21         | 4.377312  | .5439989  | 8.05   | 0.000 | 3.307678  | 5.446946  |
| FE_C22         | .440006   | .3614377  | 1.22   | 0.224 | -.2706683 | 1.15068   |
| FE_C23         | .5709749  | .2844041  | 2.01   | 0.045 | .0117674  | 1.130182  |
| FE_C24         | -.3788059 | .2696841  | -1.40  | 0.161 | -.9090703 | .1514585  |
| FE_C25         | 4.280575  | .828669   | 5.17   | 0.000 | 2.651211  | 5.90994   |
| FE_C26         | 2.414662  | .572631   | 4.22   | 0.000 | 1.288731  | 3.540594  |
| FE_C27         | 4.737185  | .699262   | 6.77   | 0.000 | 3.362266  | 6.112104  |
| FE_C28         | 5.169663  | 1.17611   | 4.40   | 0.000 | 2.857146  | 7.48218   |
| FE_C29         | 1.594082  | .1924493  | 8.28   | 0.000 | 1.21568   | 1.972484  |
| FE_C30         | 4.724302  | 1.234626  | 3.83   | 0.000 | 2.296727  | 7.151877  |
| FE_C31         | 4.339629  | .9377876  | 4.63   | 0.000 | 2.49571   | 6.183547  |
| FE_C32         | -3.164962 | .5664826  | -5.59  | 0.000 | -4.278804 | -2.051119 |
| FE_C33         | .9308979  | .1632305  | 5.70   | 0.000 | .6099471  | 1.251849  |
| FE_C34         | 1.566039  | .5218332  | 3.00   | 0.003 | .5399885  | 2.59209   |
| FE_C35         | 2.992975  | .5136192  | 5.83   | 0.000 | 1.983075  | 4.002875  |
| FE_C36         | 1.306088  | .2006215  | 6.51   | 0.000 | .9116172  | 1.700558  |
| FE_C37         | 2.768777  | .321355   | 8.62   | 0.000 | 2.136915  | 3.400639  |
| FE_C38         | 3.101544  | .6497272  | 4.77   | 0.000 | 1.824022  | 4.379065  |
| FE_C39         | 2.252395  | .1707517  | 13.19  | 0.000 | 1.916655  | 2.588134  |
| FE_C40         | -2.465774 | .5656156  | -4.36  | 0.000 | -3.577912 | -1.353636 |
| FE_C41         | 2.822566  | .490196   | 5.76   | 0.000 | 1.858722  | 3.786411  |
| FE_C42         | 2.631297  | .792021   | 3.32   | 0.001 | 1.073991  | 4.188603  |
| FE_C43         | .7973856  | .2322975  | 3.43   | 0.001 | .3406323  | 1.254139  |
| FE_C44         | 2.161841  | .32546    | 6.64   | 0.000 | 1.521908  | 2.801774  |
| FE_C45         | 2.300136  | .80.34616 | 0.03   | 0.977 | -155.6799 | 160.2802  |
| FE_C46         | -1.439658 | .6520947  | -2.21  | 0.028 | -2.721835 | -.1574816 |
| FE_C47         | 3.454958  | .9047732  | 3.82   | 0.000 | 1.675954  | 5.233962  |
| FE_C48         | -.9532801 | 158.3417  | -0.01  | 0.995 | -312.2915 | 310.3849  |
| FE_C49         | -.3967721 | 1035.421  | -0.00  | 1.000 | -2036.287 | 2035.493  |
| FE_C50         | 2.544896  | .2541699  | 10.01  | 0.000 | 2.045136  | 3.044655  |
| FE_C51         | -2.068677 | .8134709  | -2.54  | 0.011 | -3.668159 | -.4691958 |
| FE_C52         | 1.908152  | .5595285  | 3.41   | 0.001 | .8079833  | 3.008321  |
| FE_Y2          | -.0575039 | .0320547  | -1.79  | 0.074 | -.1205311 | .0055234  |
| FE_Y3          | -.022077  | .0336548  | -0.66  | 0.512 | -.0882505 | .0440965  |
| FE_Y4          | -.0428749 | .0354706  | -1.21  | 0.228 | -.1126187 | .0268689  |
| FE_Y5          | -.0487722 | .032127   | -1.52  | 0.130 | -.1119417 | .0143972  |
| FE_Y6          | .0971967  | .0374236  | 2.60   | 0.010 | .0236128  | .1707807  |
| FE_Y7          | -.0776118 | .0405229  | -1.92  | 0.056 | -.1572897 | .0020662  |
| FE_Y8          | -.1086879 | .0425108  | -2.56  | 0.011 | -.1922745 | -.0251014 |
| FE_Y9          | -.1475424 | .0440815  | -3.35  | 0.001 | -.2342174 | -.0608675 |
| _cons          | 5.245273  | .2492626  | 21.04  | 0.000 | 4.755163  | 5.735384  |

```
170 . estimates store S4_WLS_check

171 .
172 . *****
173 . * Expected key S4 point estimates from R
174 . *
175 . * Austria      about -0.504
176 . * Belgium      about -0.575
177 . * Bulgaria      about -1.781
178 . * Croatia      about -1.529
179 . * Lithuania     about -2.164
180 . * Poland        about -1.524
181 . * Romania       about -1.597
182 . * Slovenia      about -1.177
183 . * Sweden        about -0.927
184 . *
185 . * Control coefficients in S4
186 . * c_log_GDPpc    about 0.376
187 . * c_log_Pop15_64 about 1.617
188 . * c_YouthUnempl  about 0.002
189 . *****
190 .
    end of do-file

191 .
```